

BFS Traversal

Step 1: Start at A (Seeking target node K)

Legend



Complete



Processing



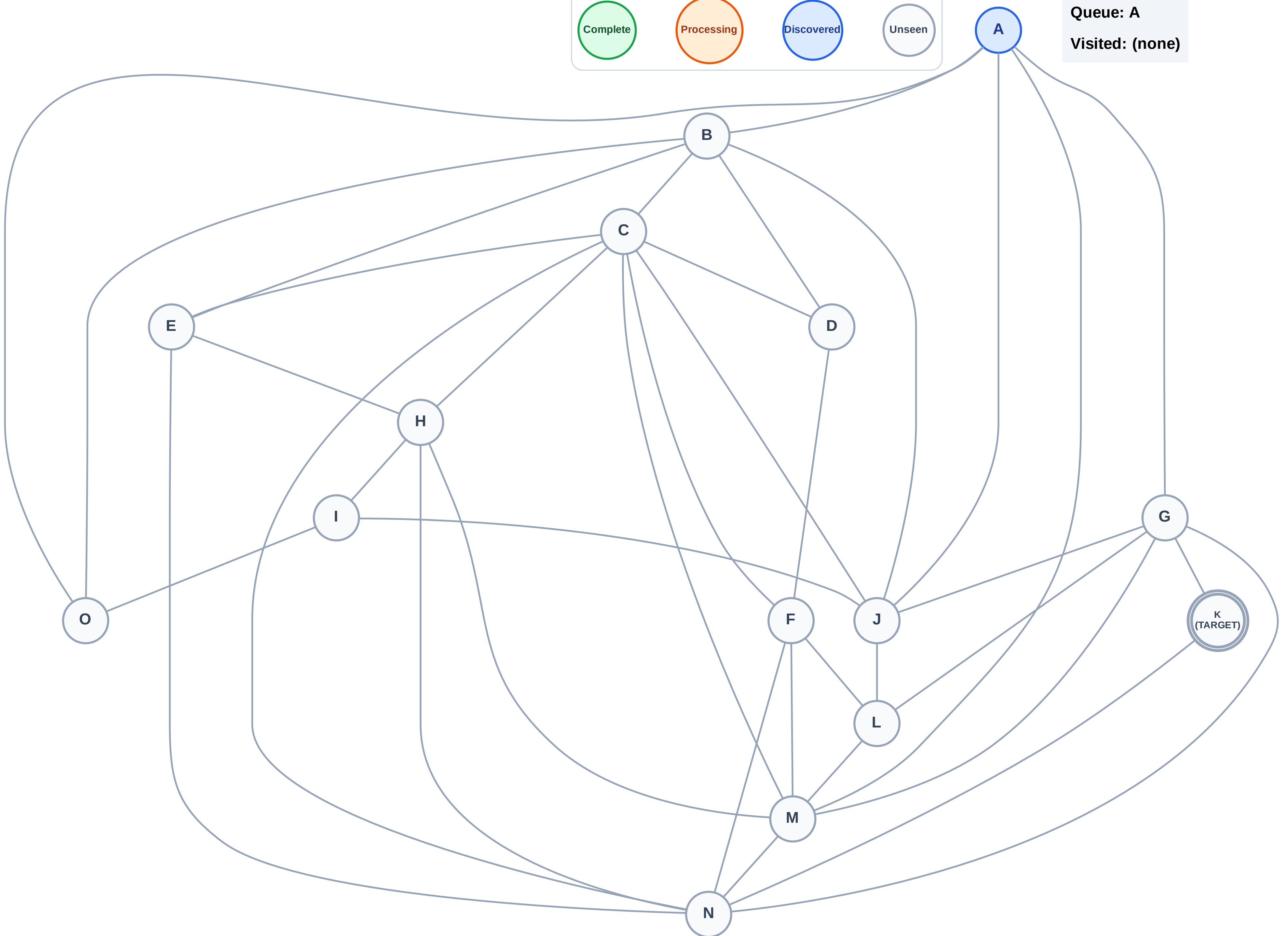
Discovered



Unseen

Queue: A

Visited: (none)



BFS Traversal

Step 2: Process node A

Legend

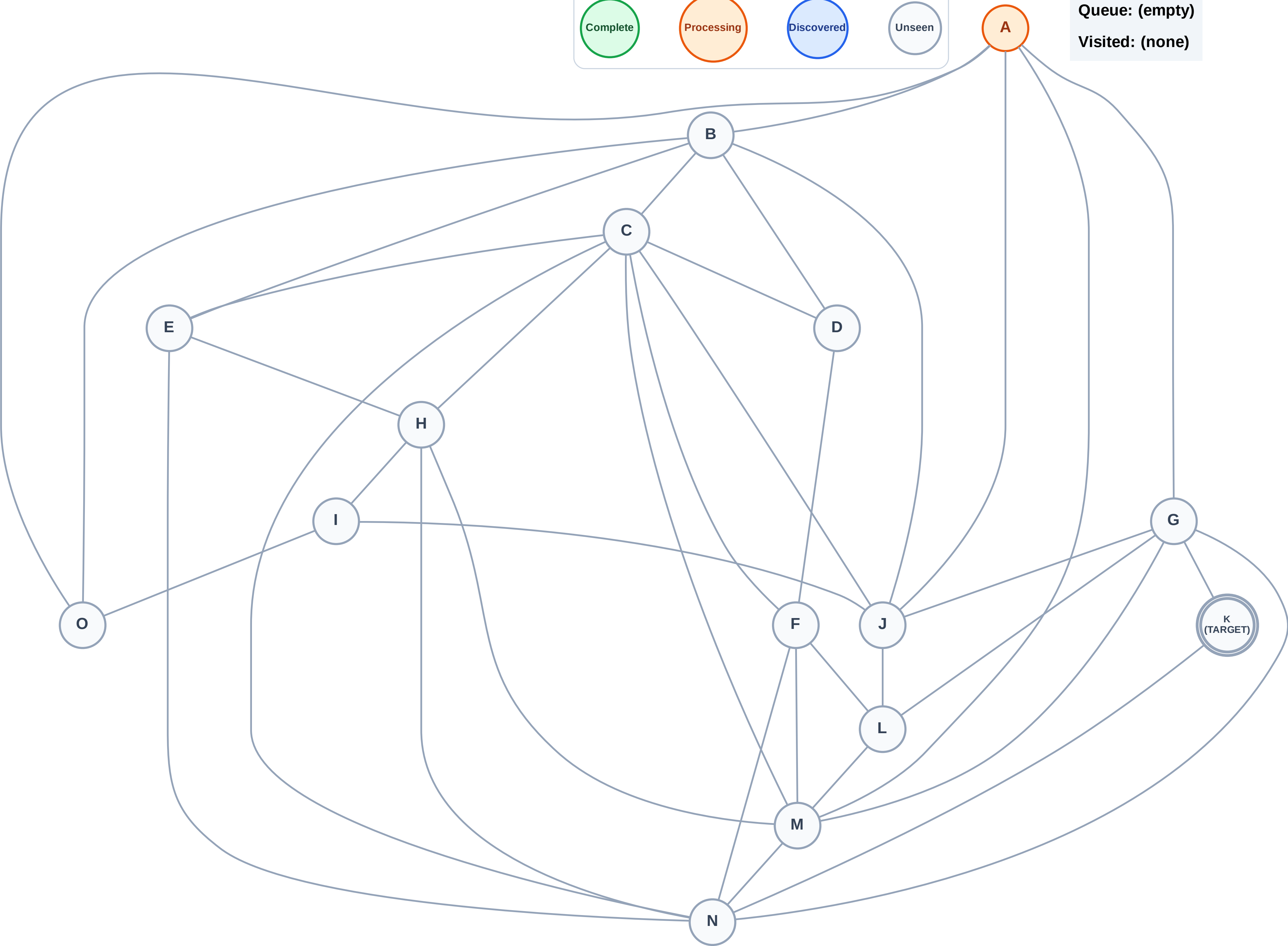
Complete

Processing

Discovered

Unseen

Queue: (empty)
Visited: (none)



BFS Traversal

Step 3: Discover B, G, J, M, O from A

Legend



Complete



Processing



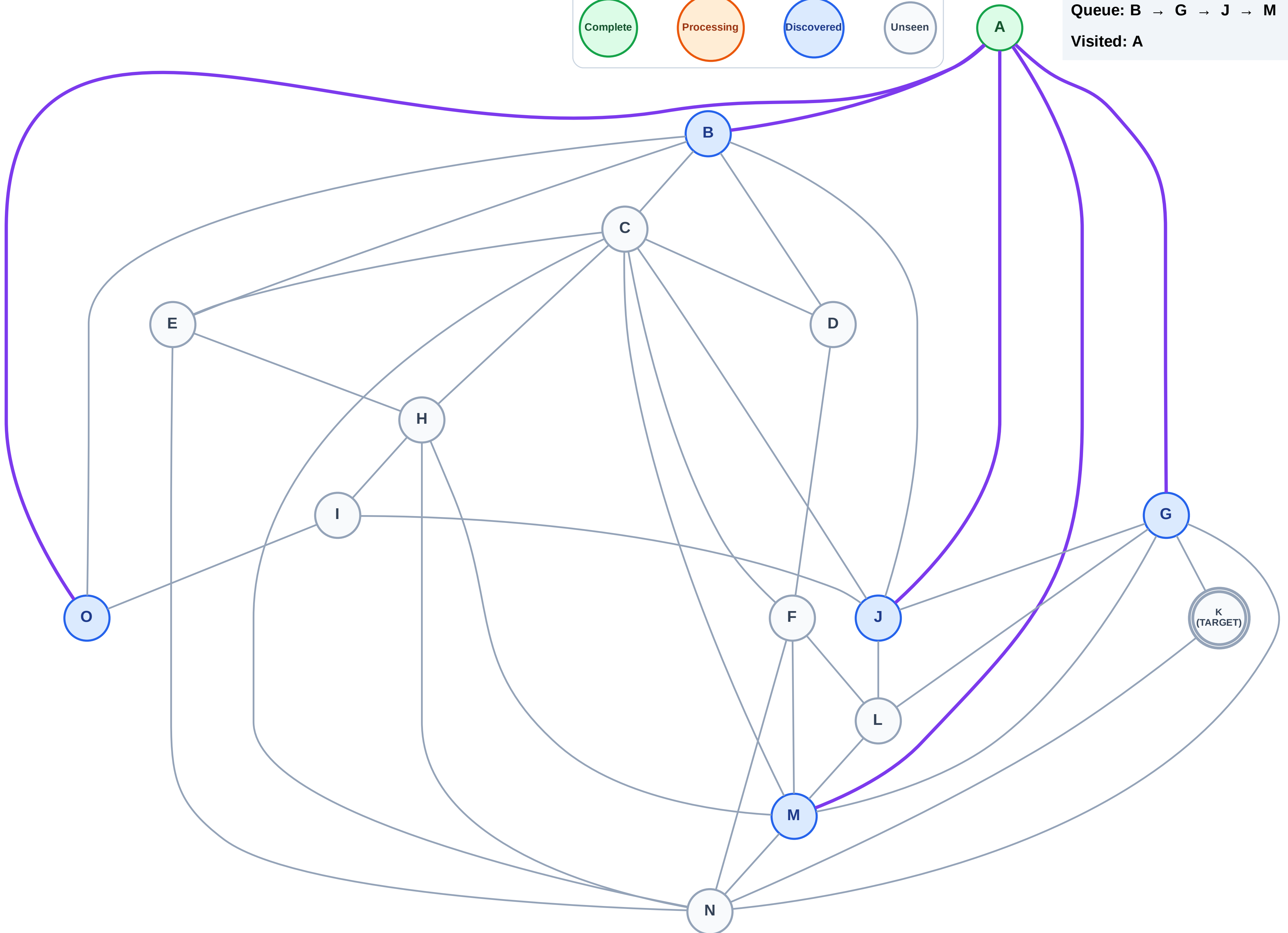
Discovered



Unseen

Queue: B → G → J → M → O

Visited: A



BFS Traversal

Step 4: Process node B

Legend

Complete

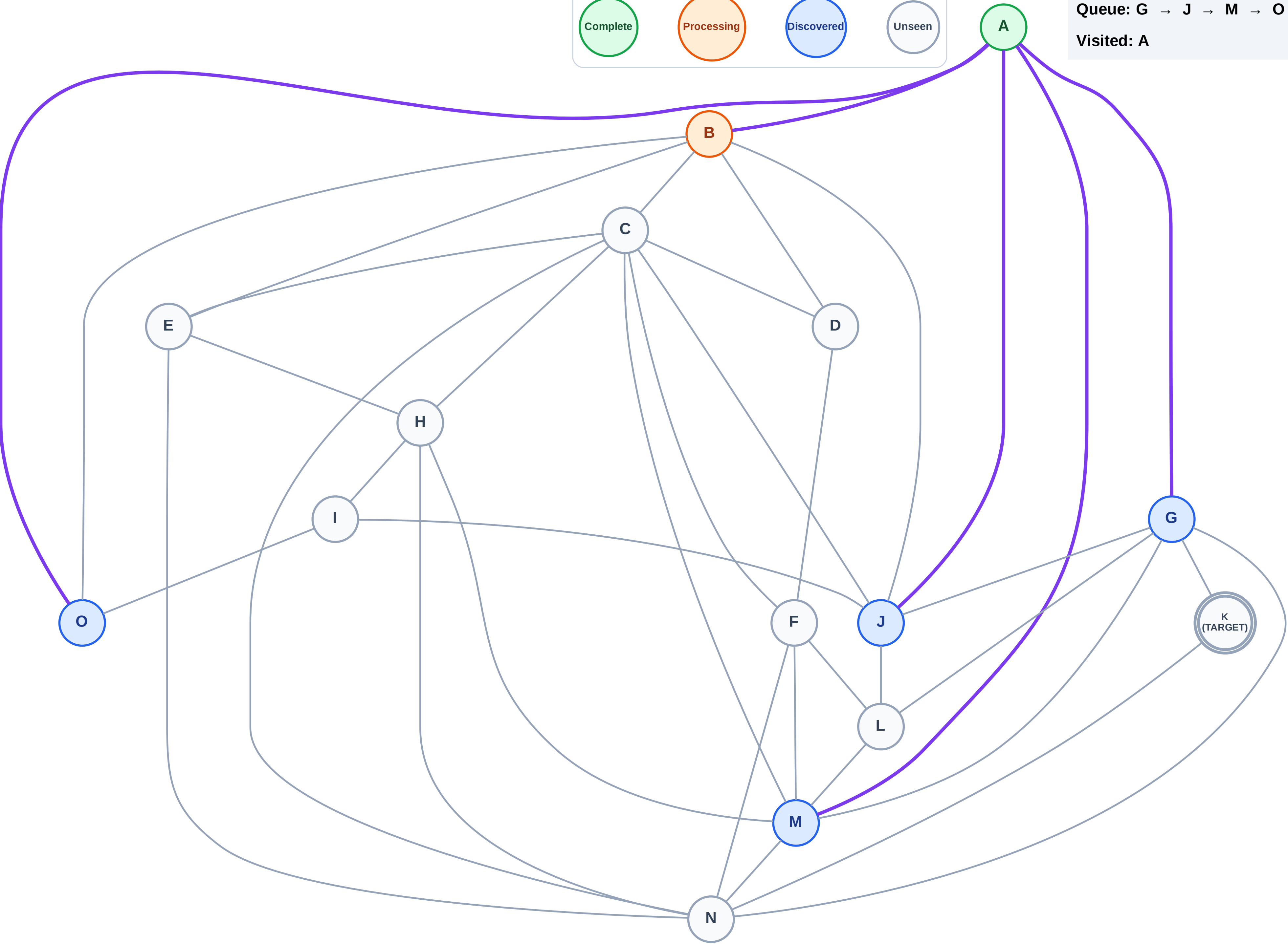
Processing

Discovered

Unseen

Queue: G → J → M → O

Visited: A



BFS Traversal

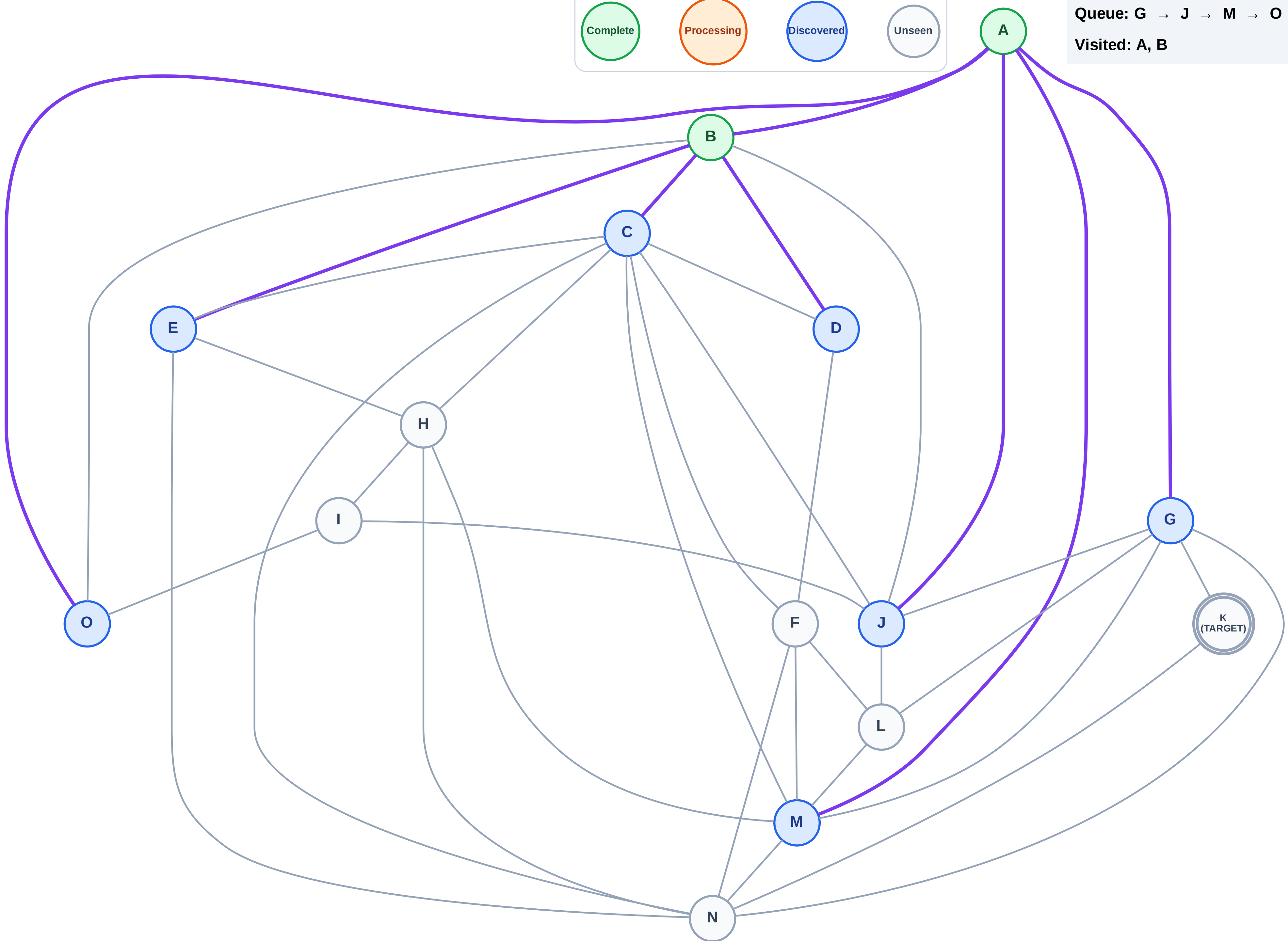
Step 5: Discover C, D, E from B

Legend



Queue: G → J → M → O → C → D → E

Visited: A, B



BFS Traversal

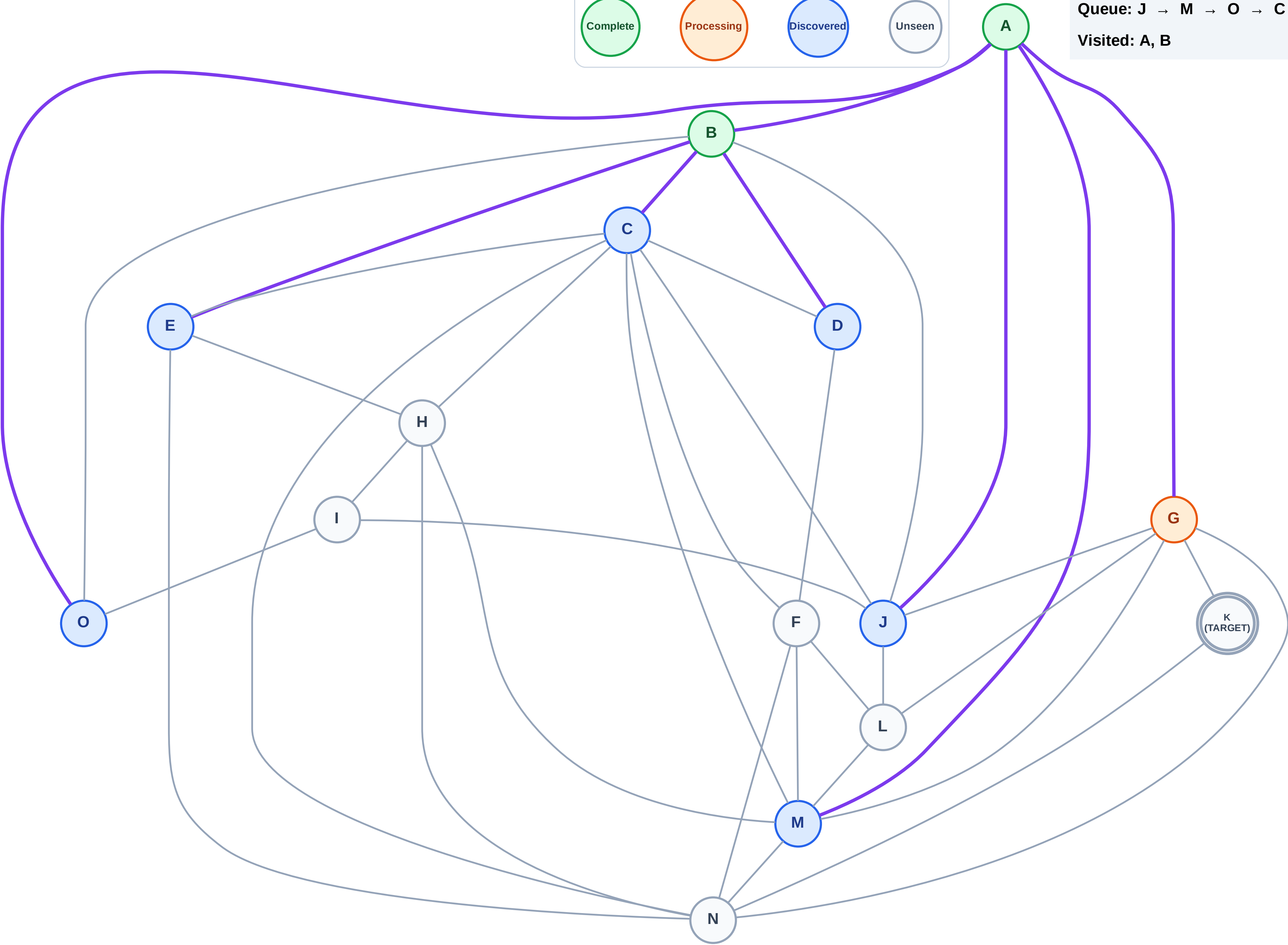
Step 6: Process node G

Legend



Queue: J → M → O → C → D → E

Visited: A, B



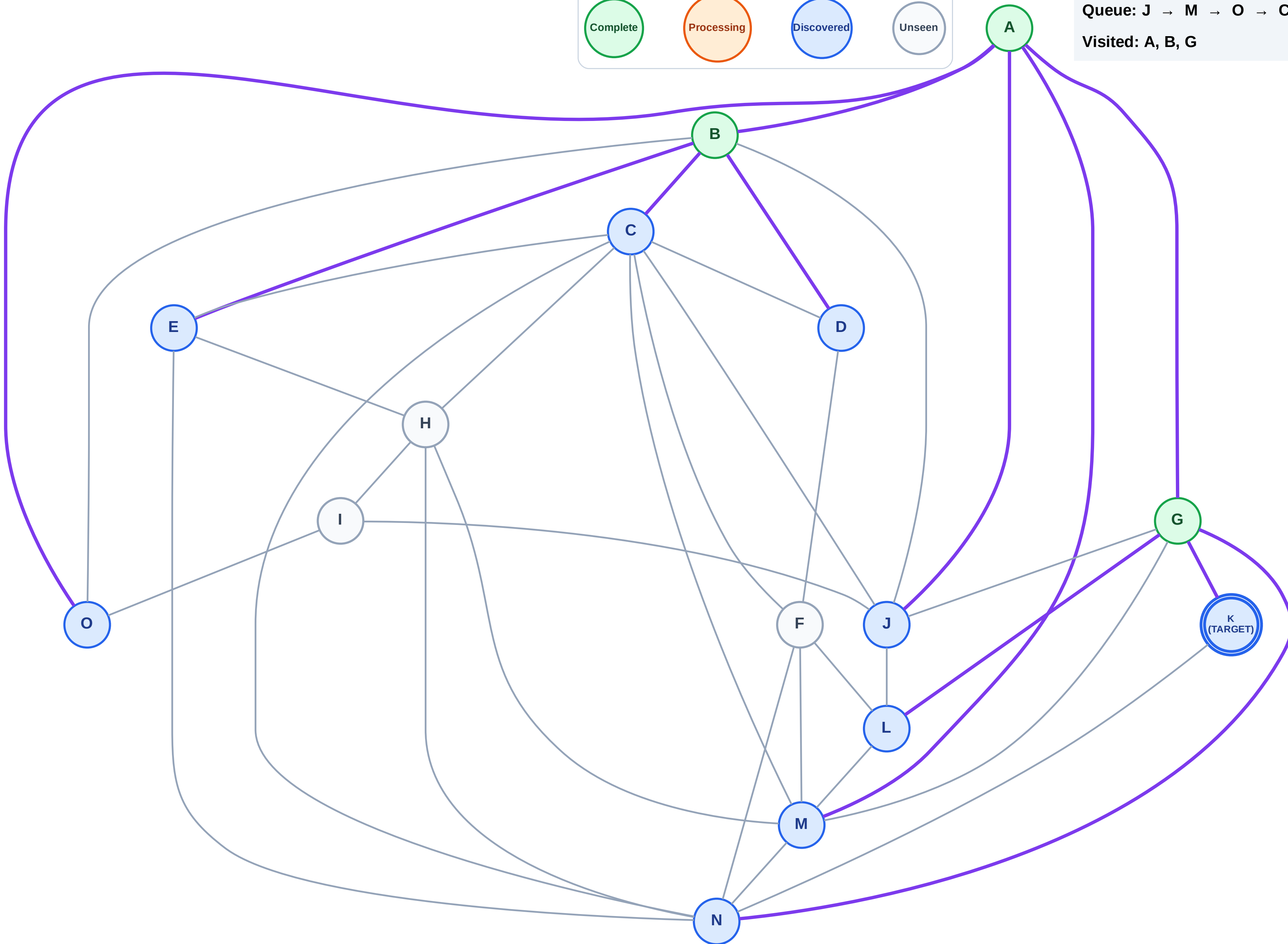
BFS Traversal

Step 7: Discover K, L, N from G

Legend



Queue: J → M → O → C → D → E → K → L → N
Visited: A, B, G



BFS Traversal

Step 8: Process node J

Legend

Complete

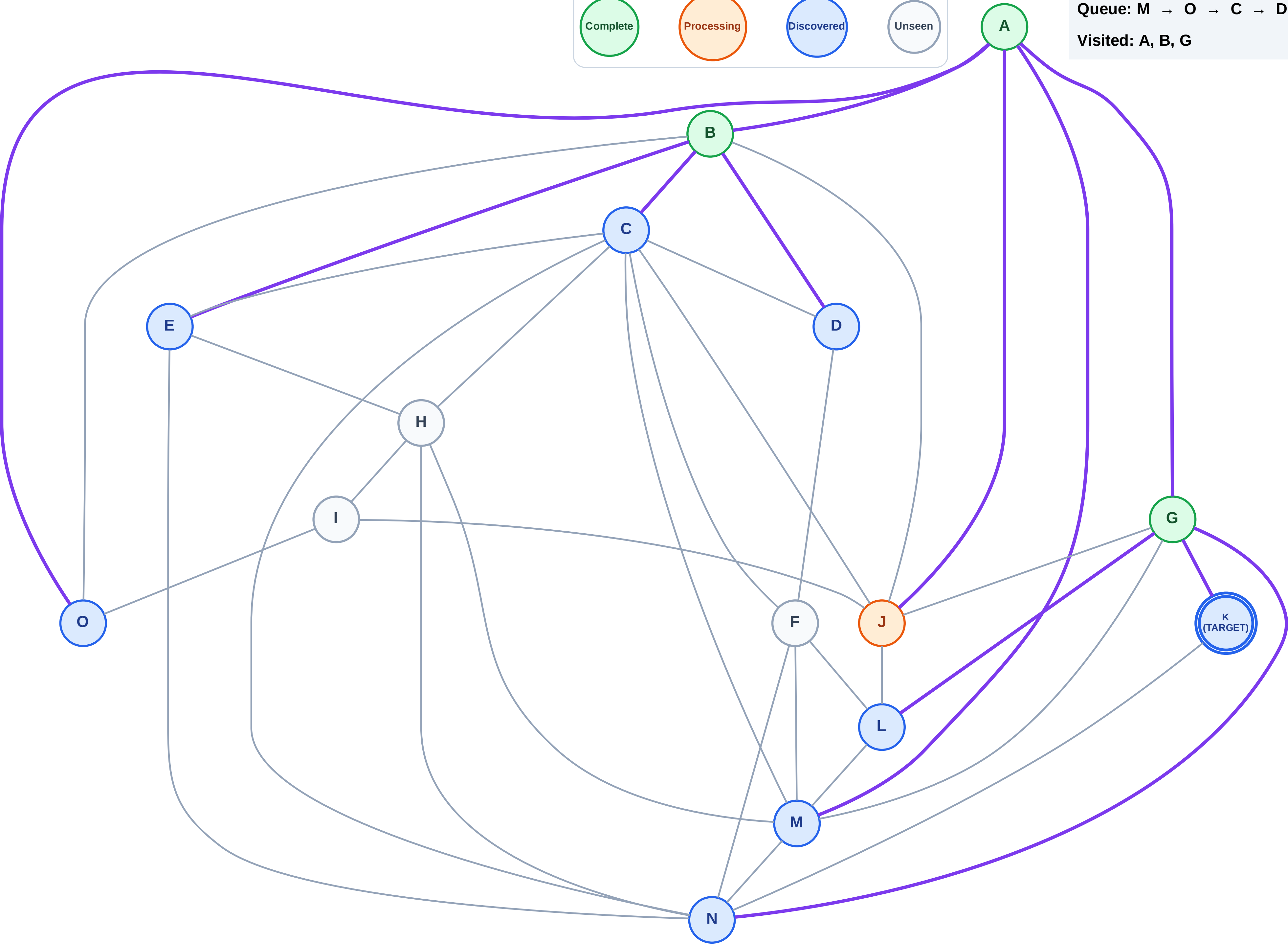
Processing

Discovered

Unseen

Queue: M → O → C → D → E → K → L → N

Visited: A, B, G



Step 9: Discover I from J

Legend

The diagram illustrates the four stages of the data science process, each represented by a colored circle:

- Complete** (Green circle)
- Processing** (Orange circle)
- Discovered** (Blue circle)
- Unseen** (Grey circle)

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- Unseen** (Grey circle)

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- Complete** (Green circle)
- Processing** (Orange circle)
- Discovered** (Blue circle)
- Unseen** (Grey circle)

The diagram illustrates the four stages of a data pipeline, each represented by a colored circle with a corresponding label:

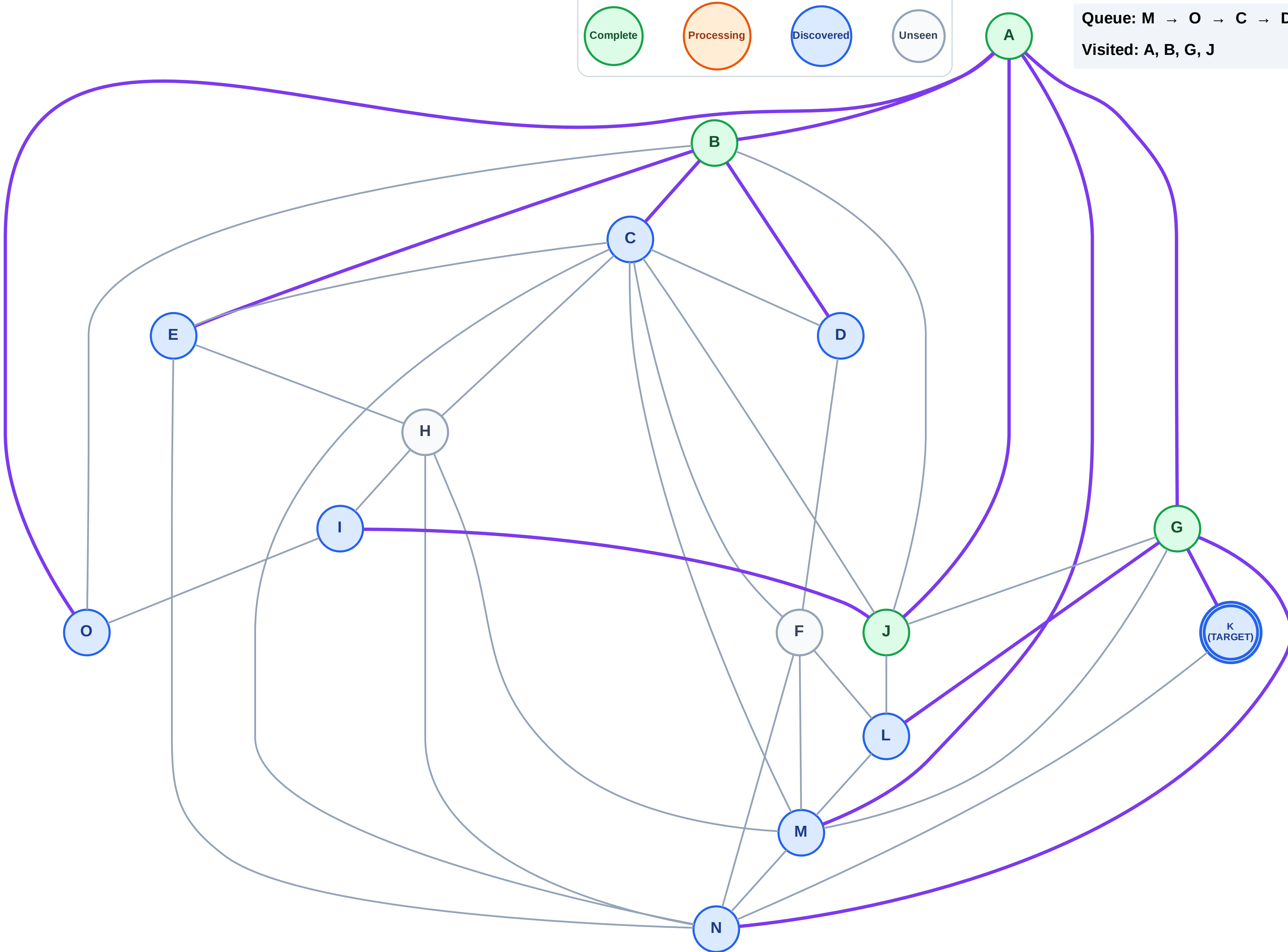
- Complete** (Green circle)
- Processing** (Orange circle)
- Discovered** (Blue circle)
- Unseen** (Grey circle)

Queue: M → O → C → D → E → K → L → N → I

Visited: A, B, G, J

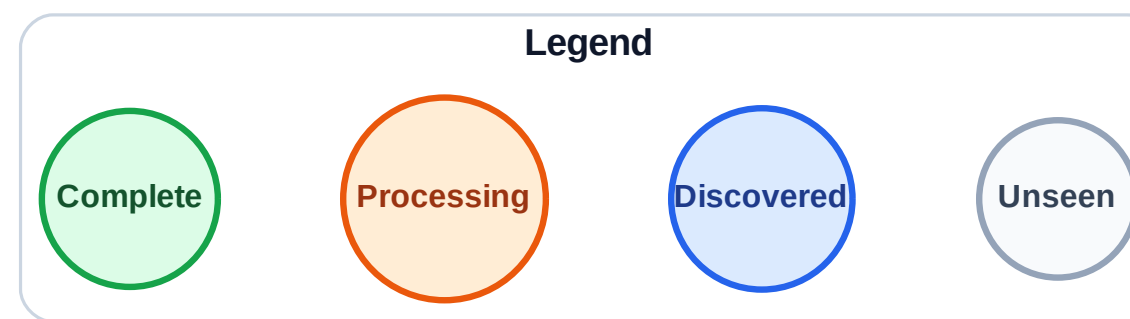
Queue: M → O → C → D → E → K → L → N → I

Visited: A, B, G, J



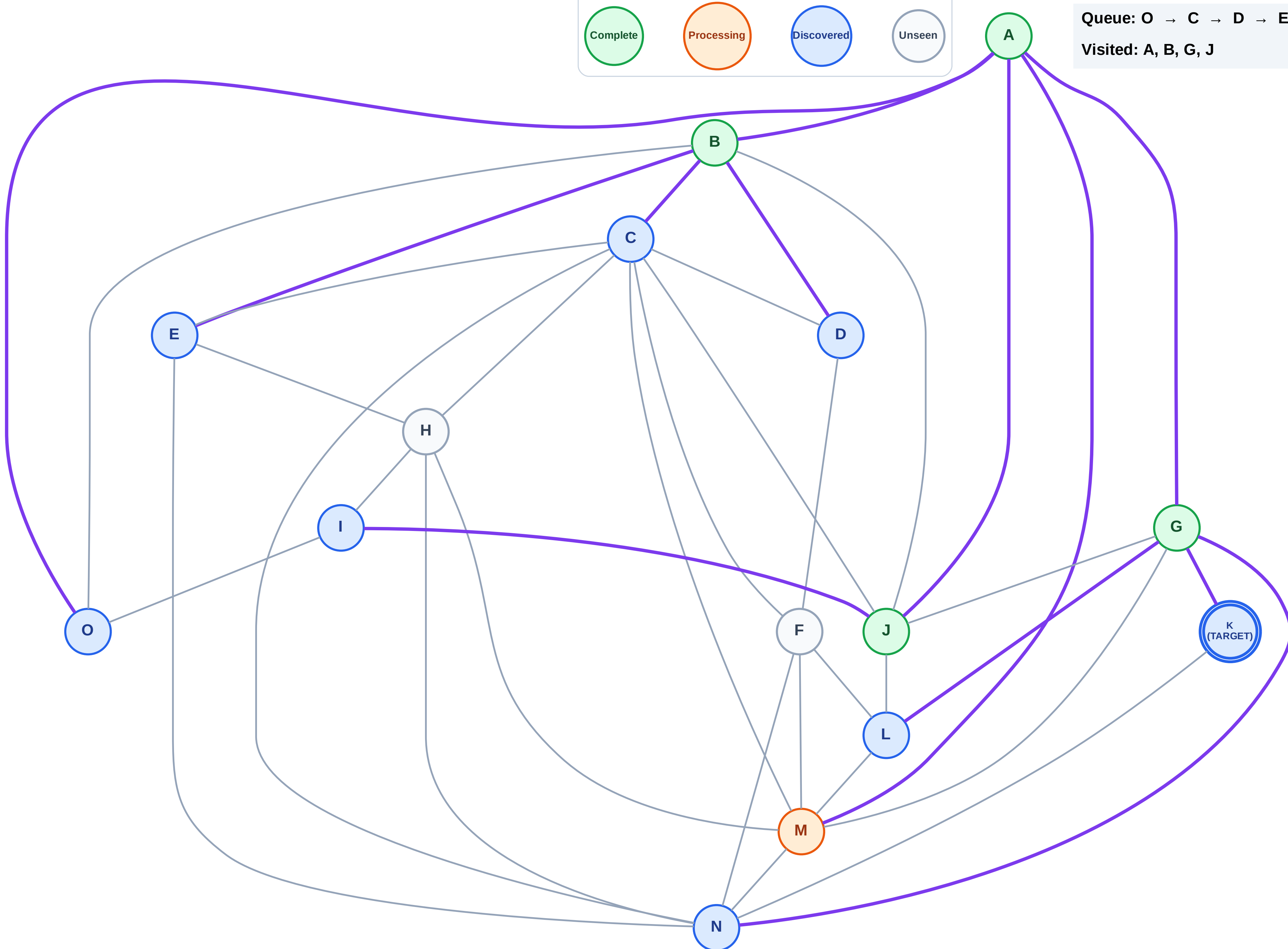
BFS Traversal

Step 10: Process node M



Queue: O → C → D → E → K → L → N → I

Visited: A, B, G, J



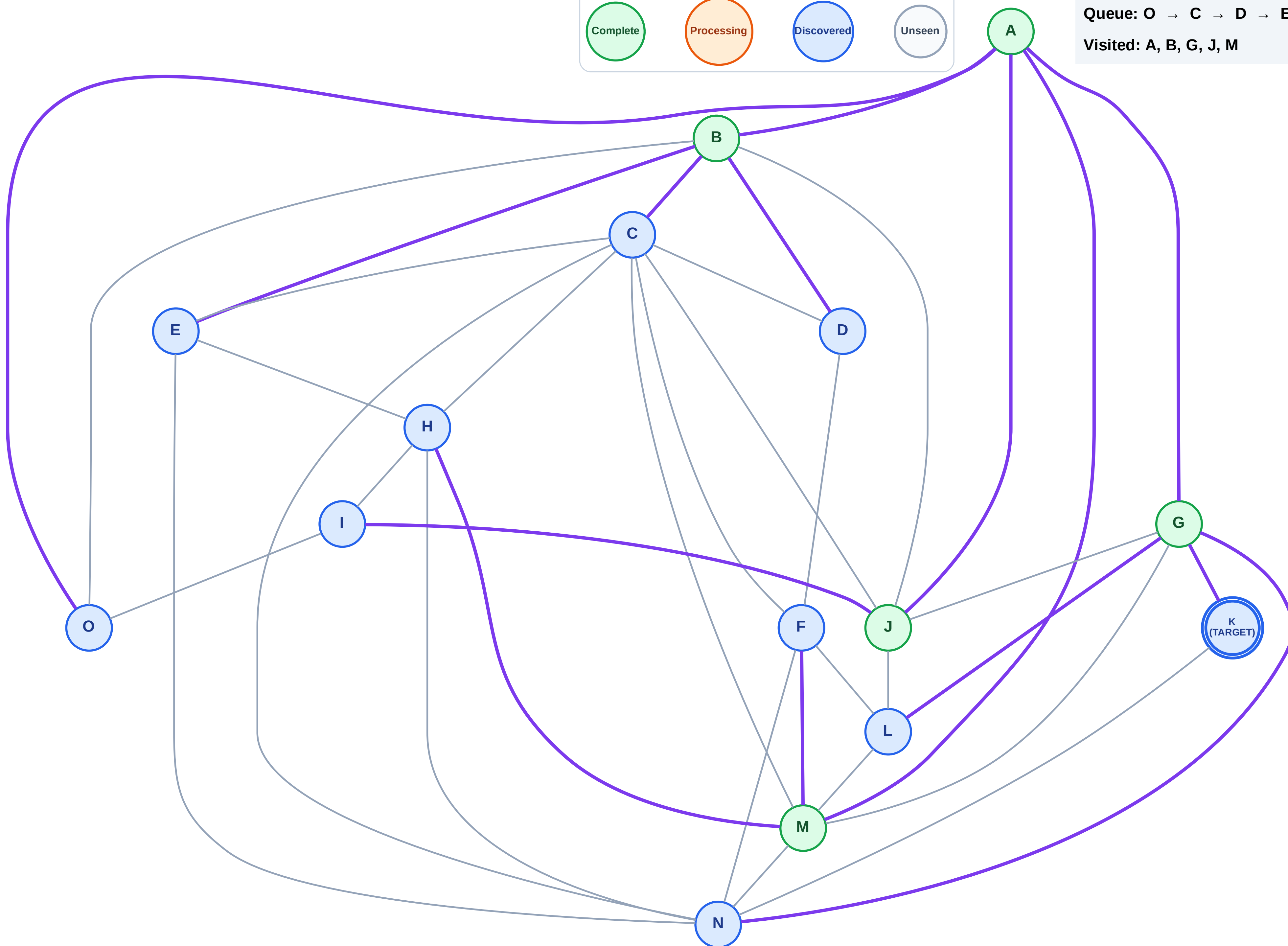
BFS Traversal

Step 11: Discover F, H from M

Legend



Queue: O → C → D → E → K → L → N → I → F → H
Visited: A, B, G, J, M



Step 12: Process node O

Legend

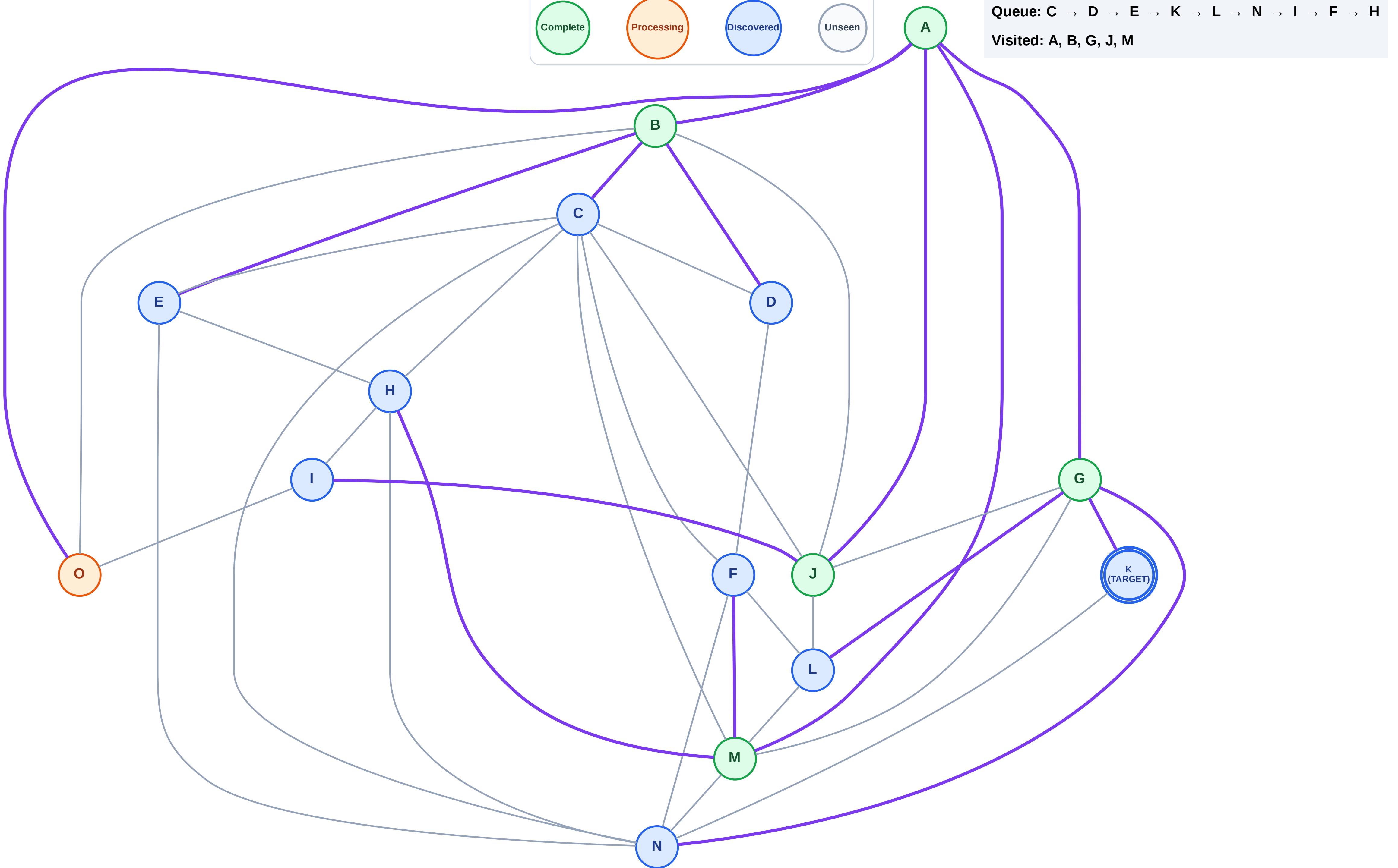
The diagram consists of four circles arranged horizontally, each containing a label. From left to right: a green circle labeled 'Complete', an orange circle labeled 'Processing', a blue circle labeled 'Discovered', and a grey circle labeled 'Unseen'. The circles are connected by a light blue line that runs along the bottom of the 'Complete', 'Processing', and 'Discovered' circles, ending under the 'Unseen' circle.

Processing

Unseen

Queue: C → D → E → K → L → N → I → F → H

Visited: A, B, G, J, M



BFS Traversal

Step 13: No new neighbors from O

Legend



Complete



Processing



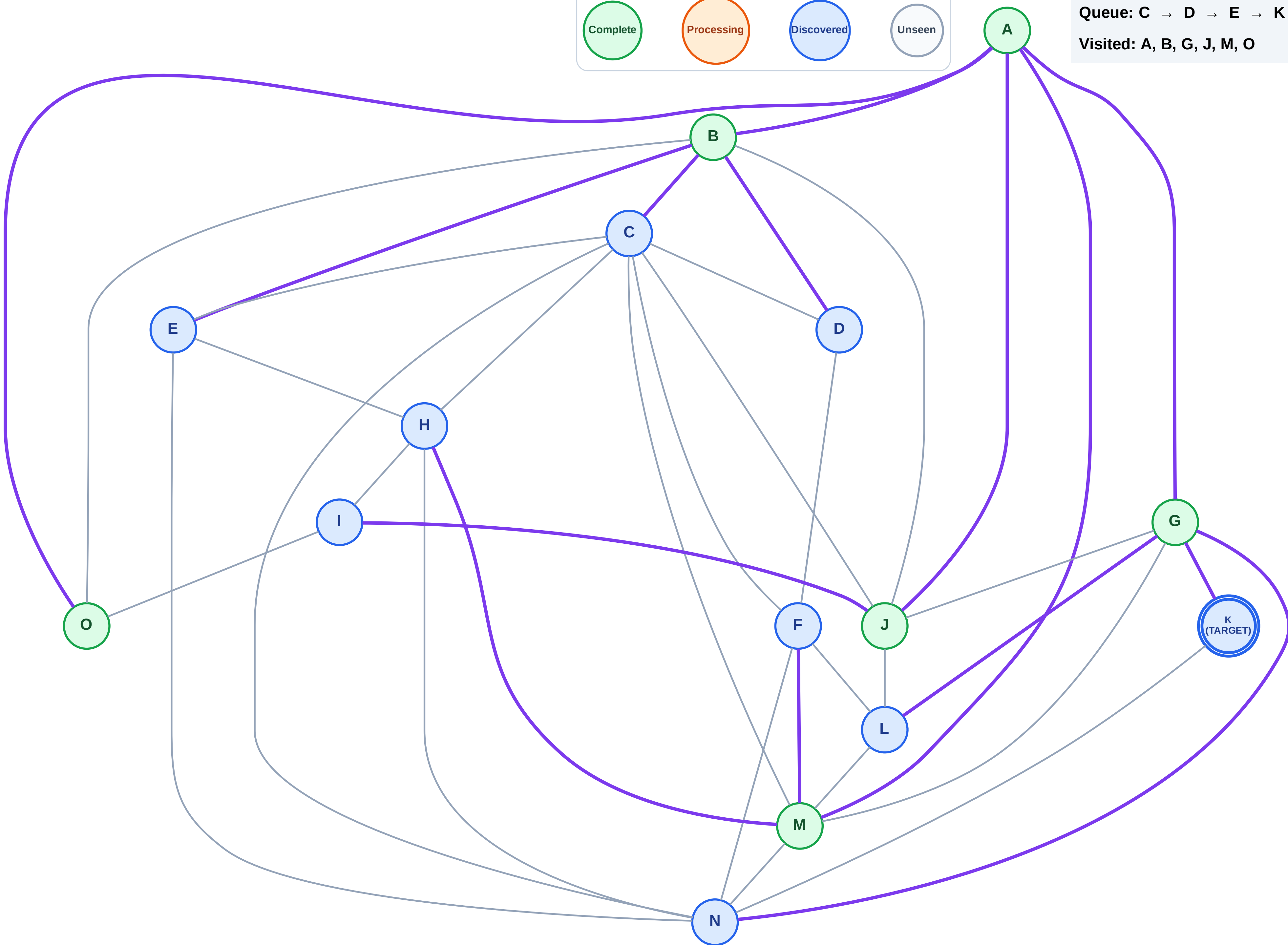
Discovered



Unseen

Queue: C → D → E → K → L → N → I → F → H

Visited: A, B, G, J, M, O



BFS Traversal

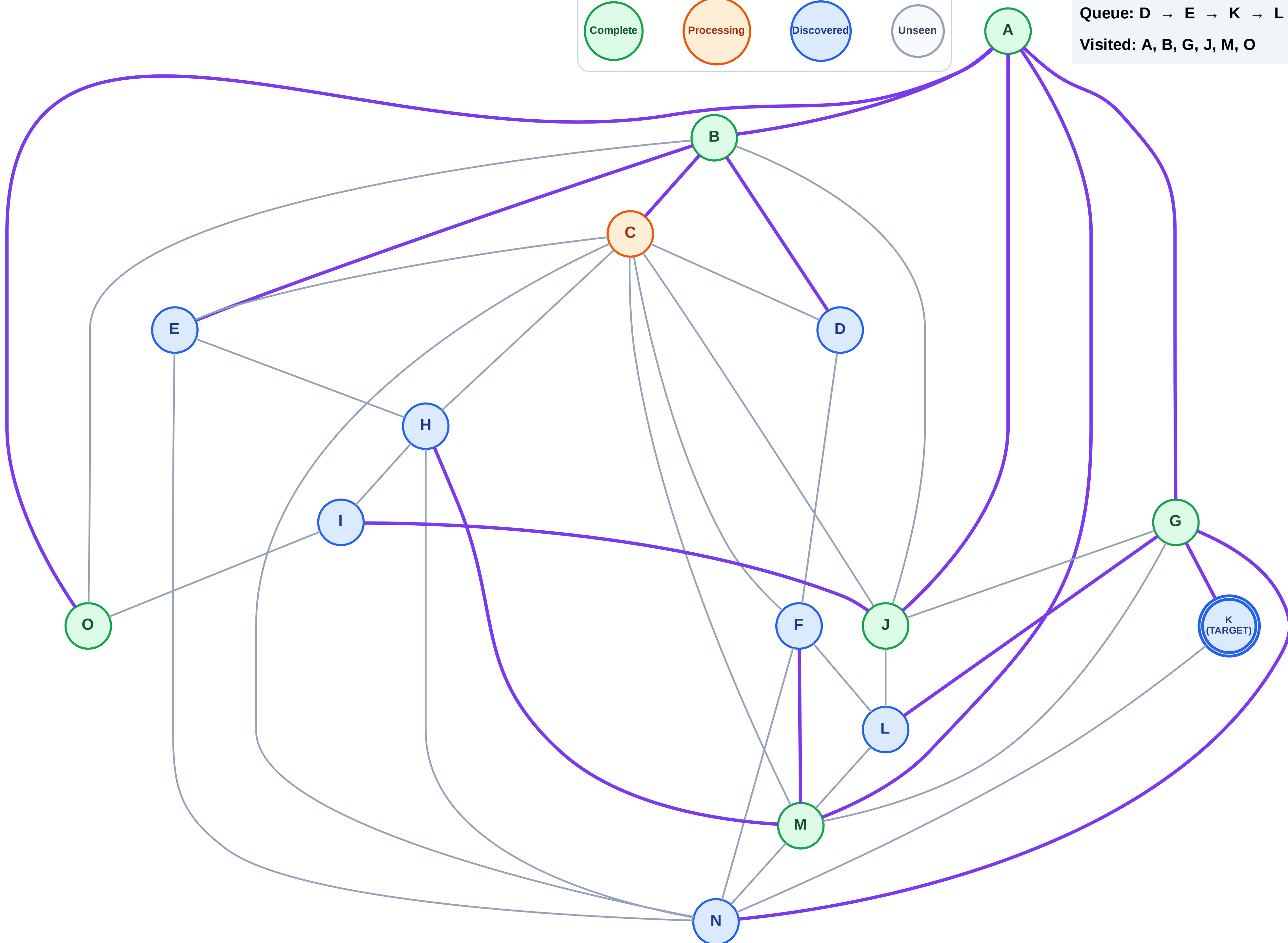
Step 14: Process node C

Legend



Queue: D → E → K → L → N → I → F → H

Visited: A, B, G, J, M, O



BFS Traversal

Step 15: No new neighbors from C

Legend



Complete



Processing

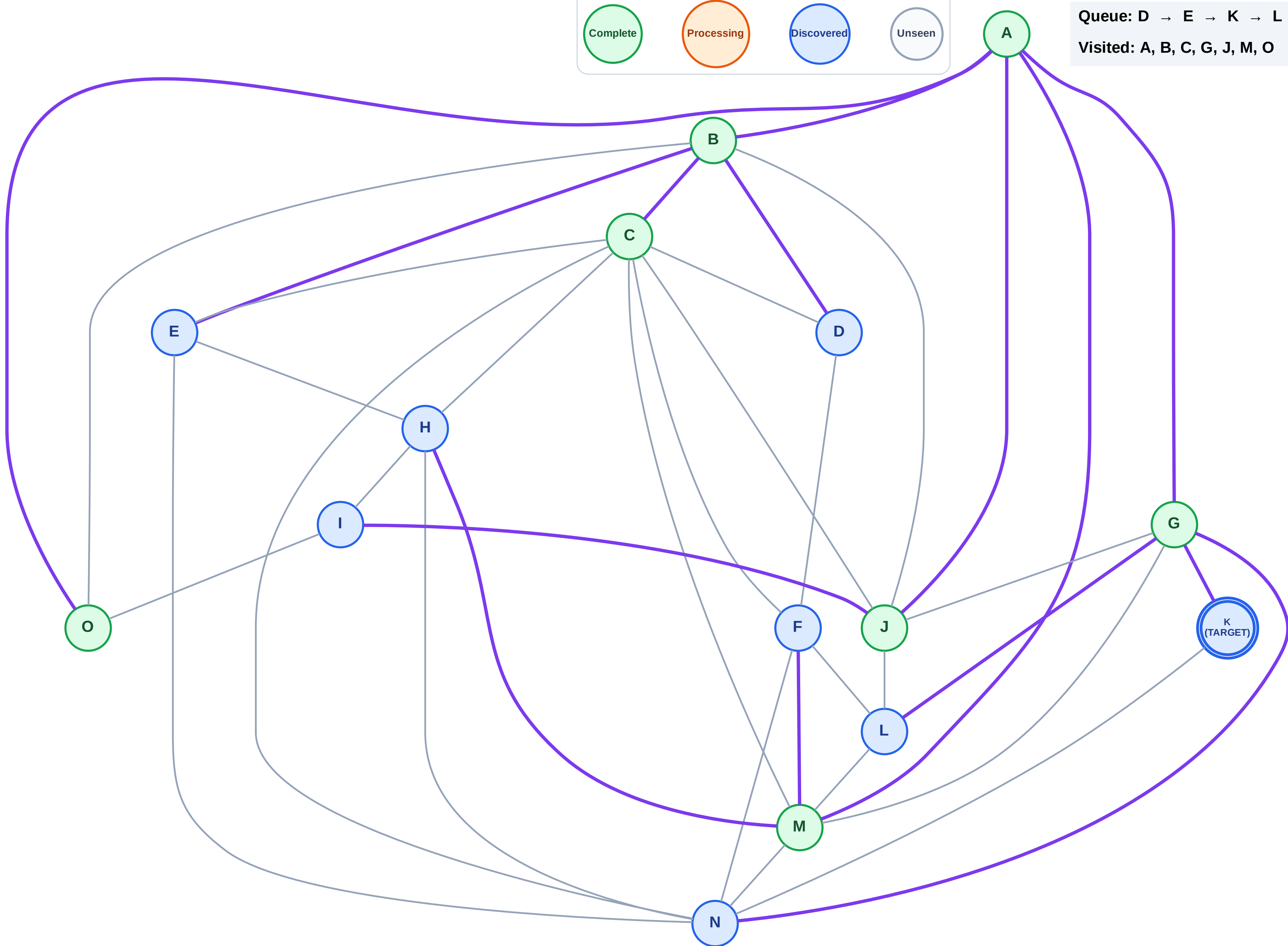


Discovered



Unseen

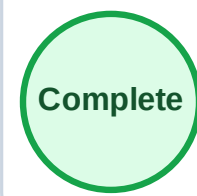
Queue: D → E → K → L → N → I → F → H
Visited: A, B, C, G, J, M, O



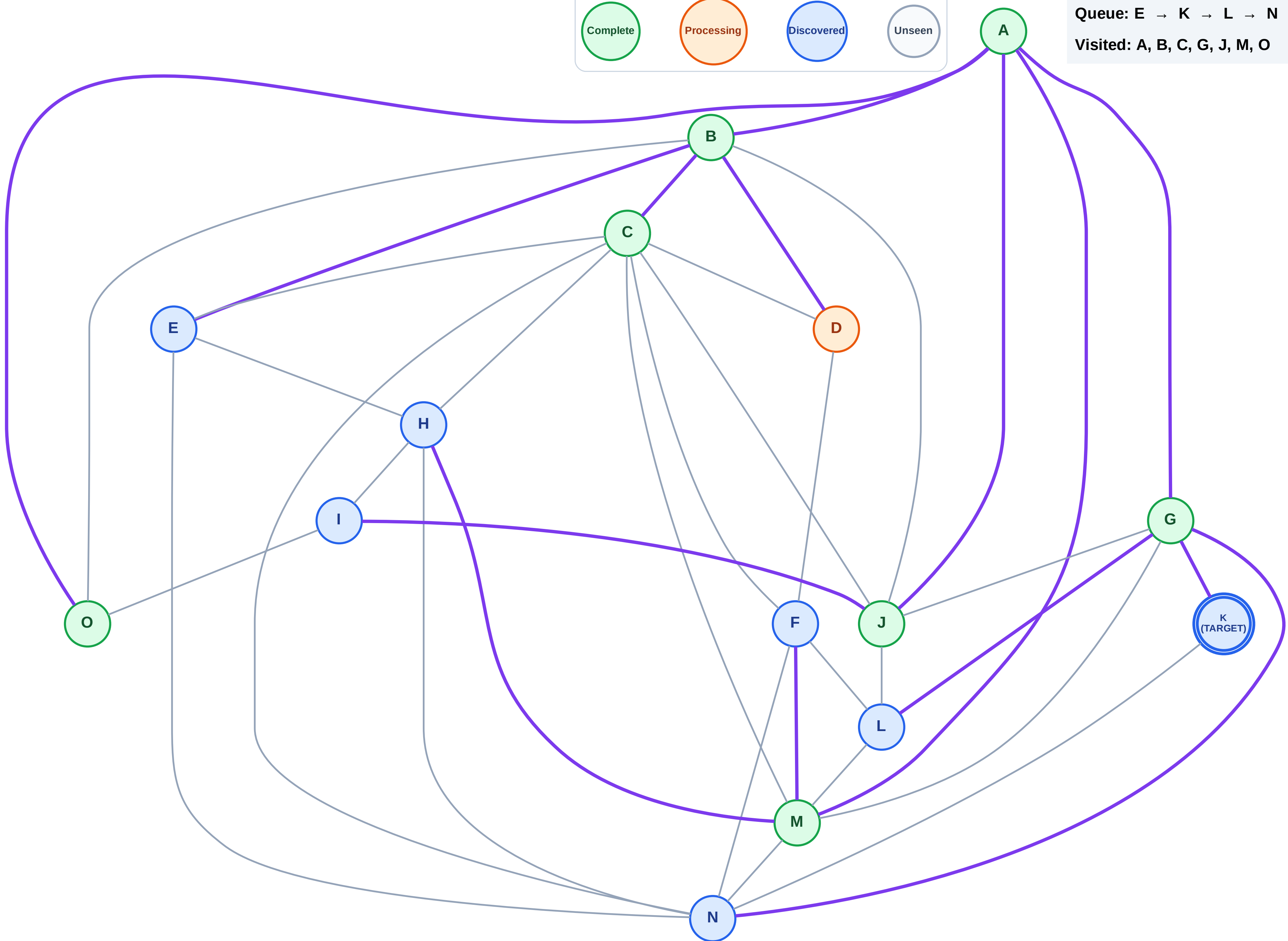
BFS Traversal

Step 16: Process node D

Legend



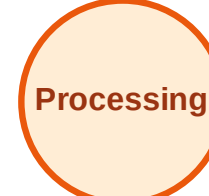
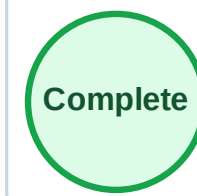
Queue: E → K → L → N → I → F → H
Visited: A, B, C, G, J, M, O



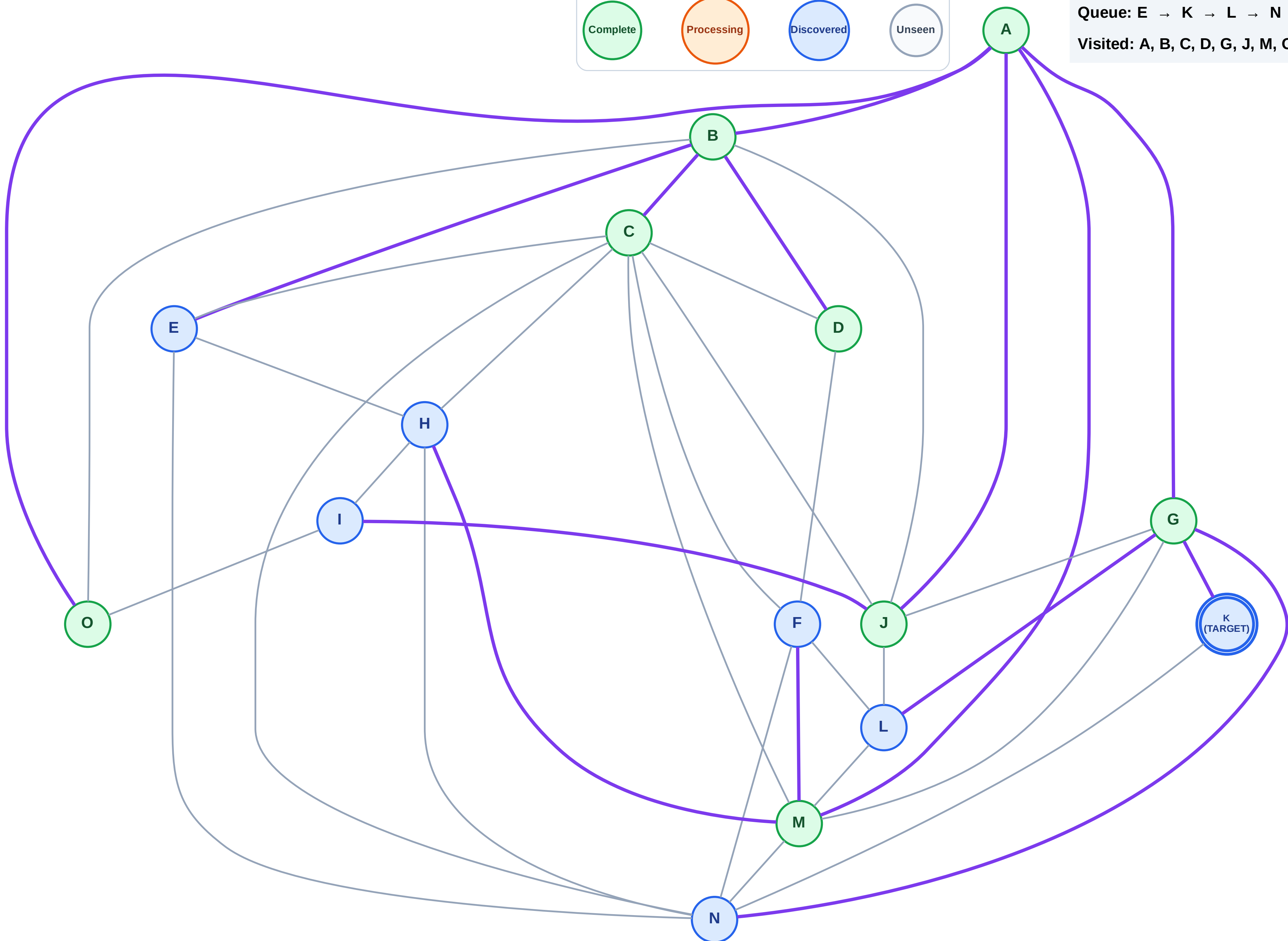
BFS Traversal

Step 17: No new neighbors from D

Legend



Queue: E → K → L → N → I → F → H
Visited: A, B, C, D, G, J, M, O



BFS Traversal

Step 18: Process node E

Legend

Complete

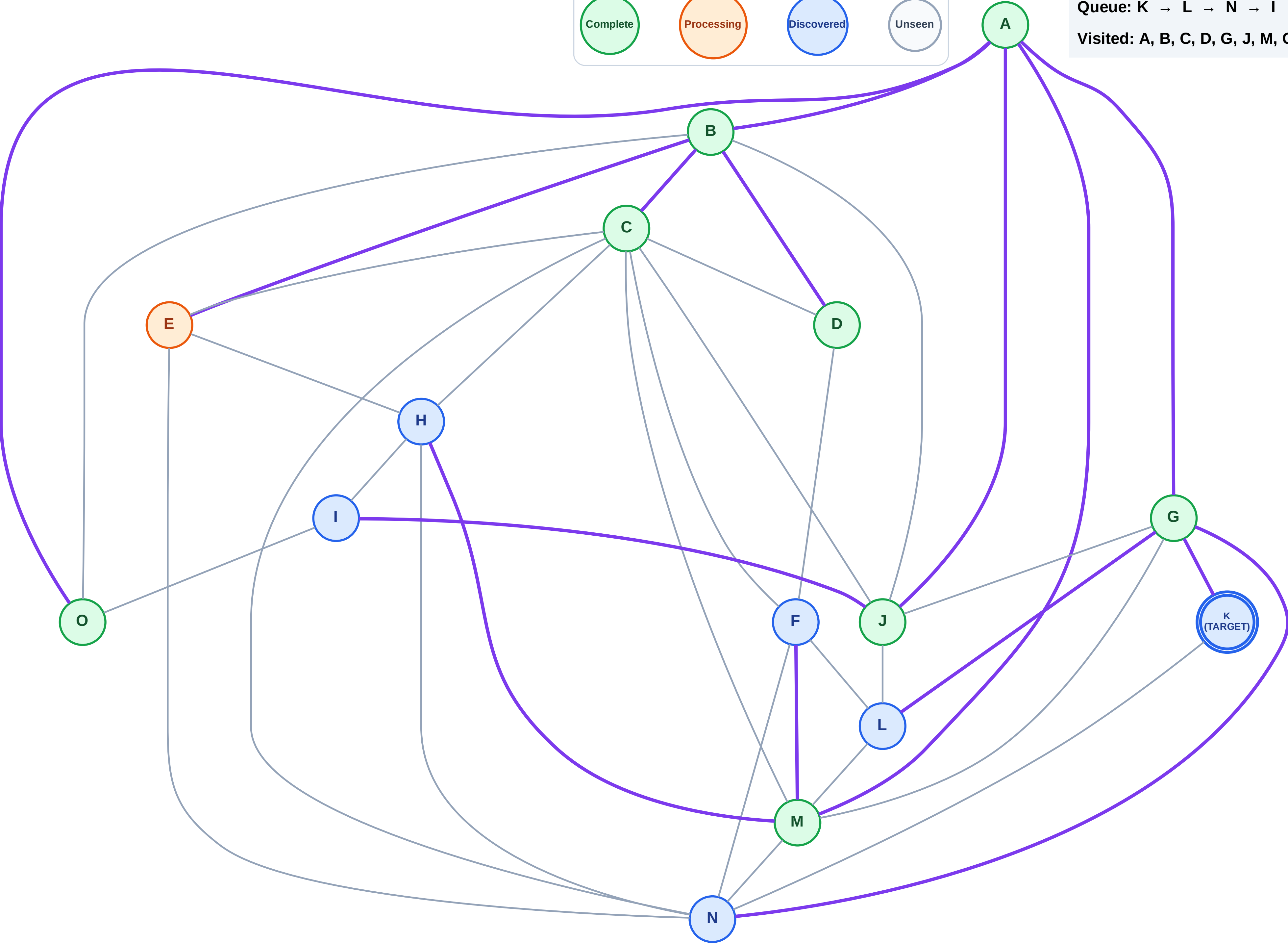
Processing

Discovered

Unseen

Queue: K → L → N → I → F → H

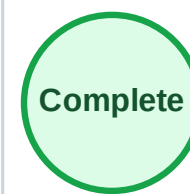
Visited: A, B, C, D, G, J, M, O



BFS Traversal

Step 19: No new neighbors from E

Legend



Complete



Processing



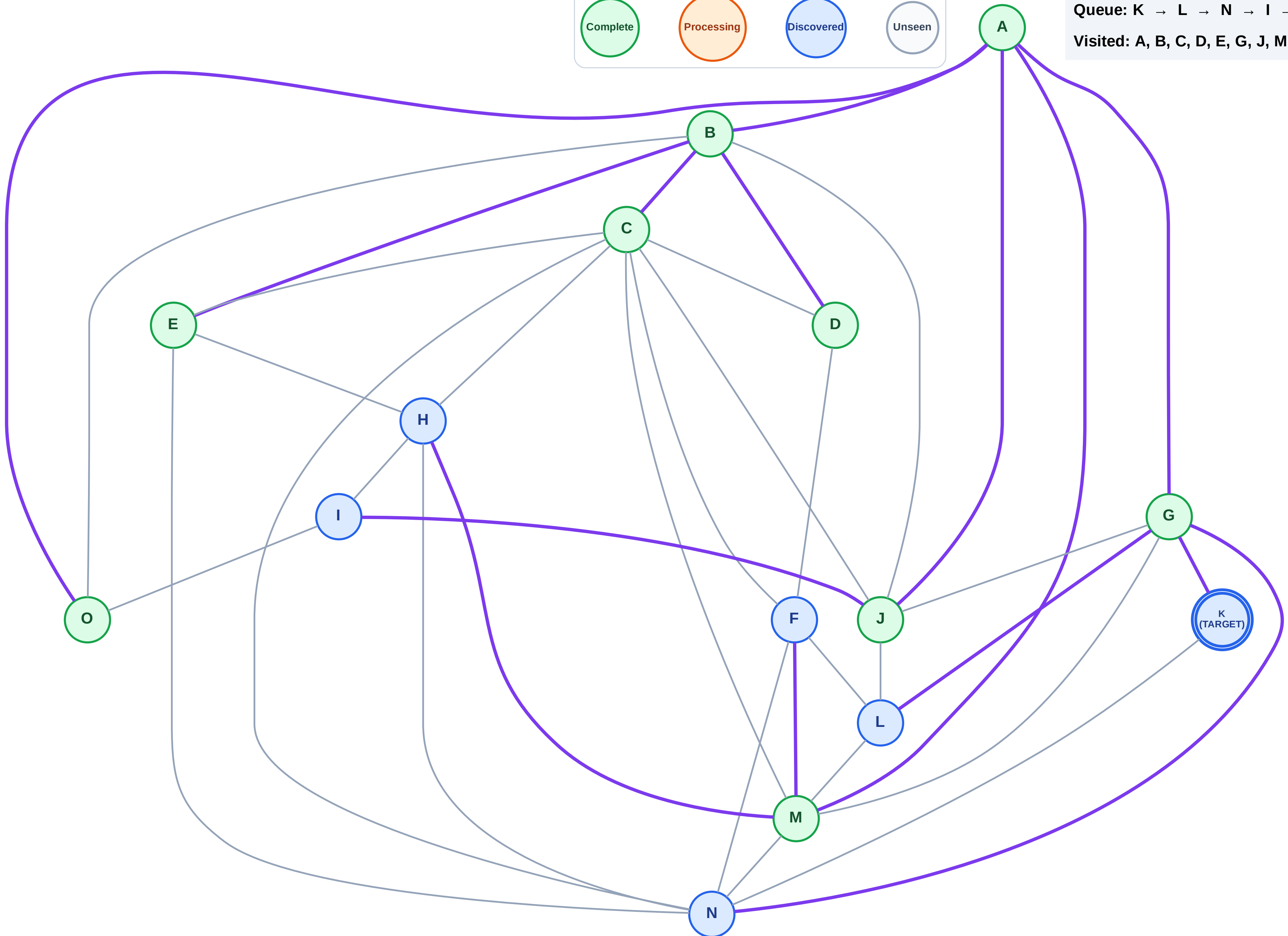
Discovered



Unseen

Queue: K → L → N → I → F → H

Visited: A, B, C, D, E, G, J, M, O



BFS Traversal

Step 20: Process node K

Legend



Complete



Processing



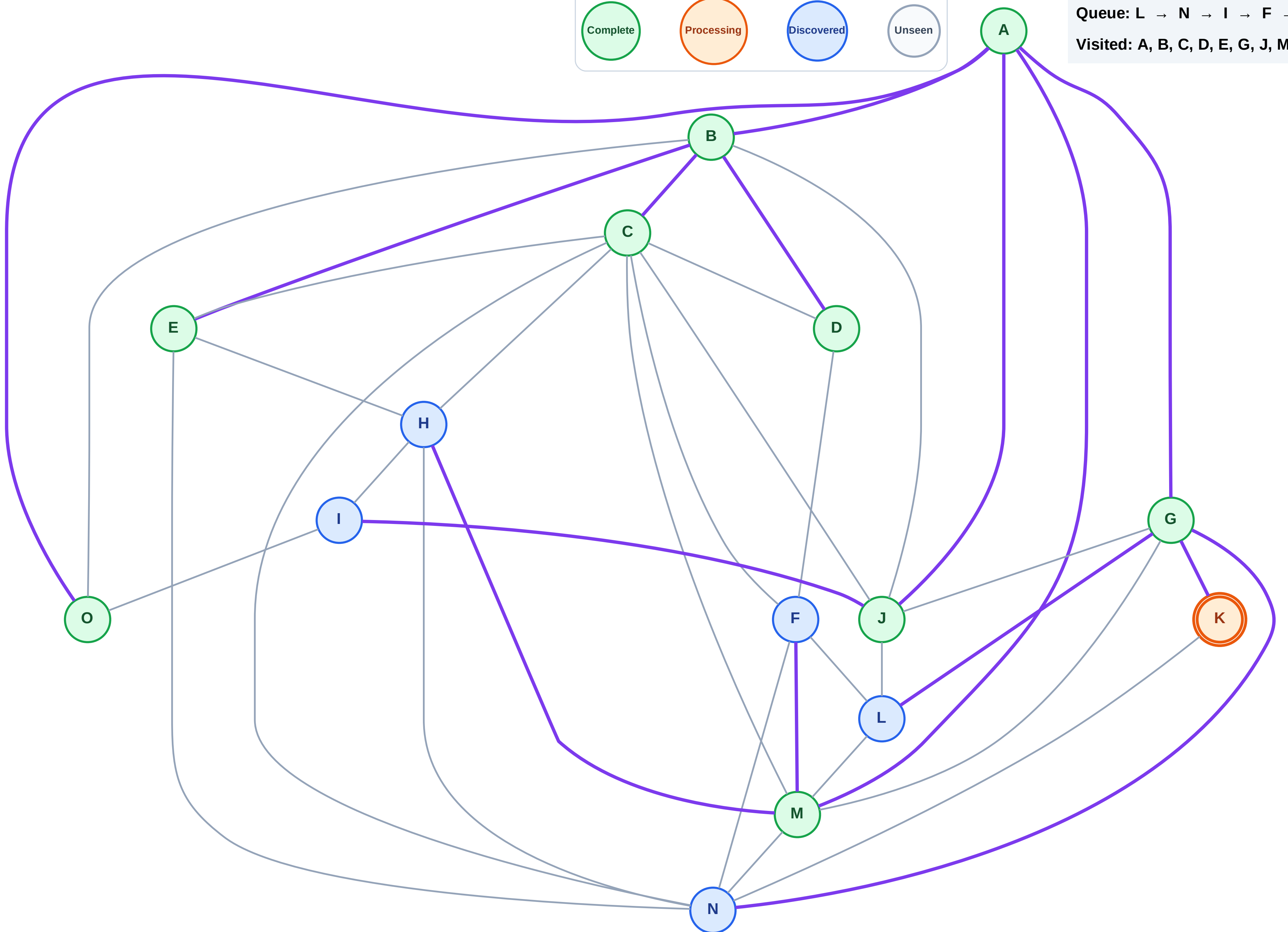
Discovered



Unseen

Queue: L → N → I → F → H

Visited: A, B, C, D, E, G, J, M, O



BFS Traversal

Step 21: TARGET NODE K REACHED! Path (2 hops): A → G → K

Legend



Complete



Processing



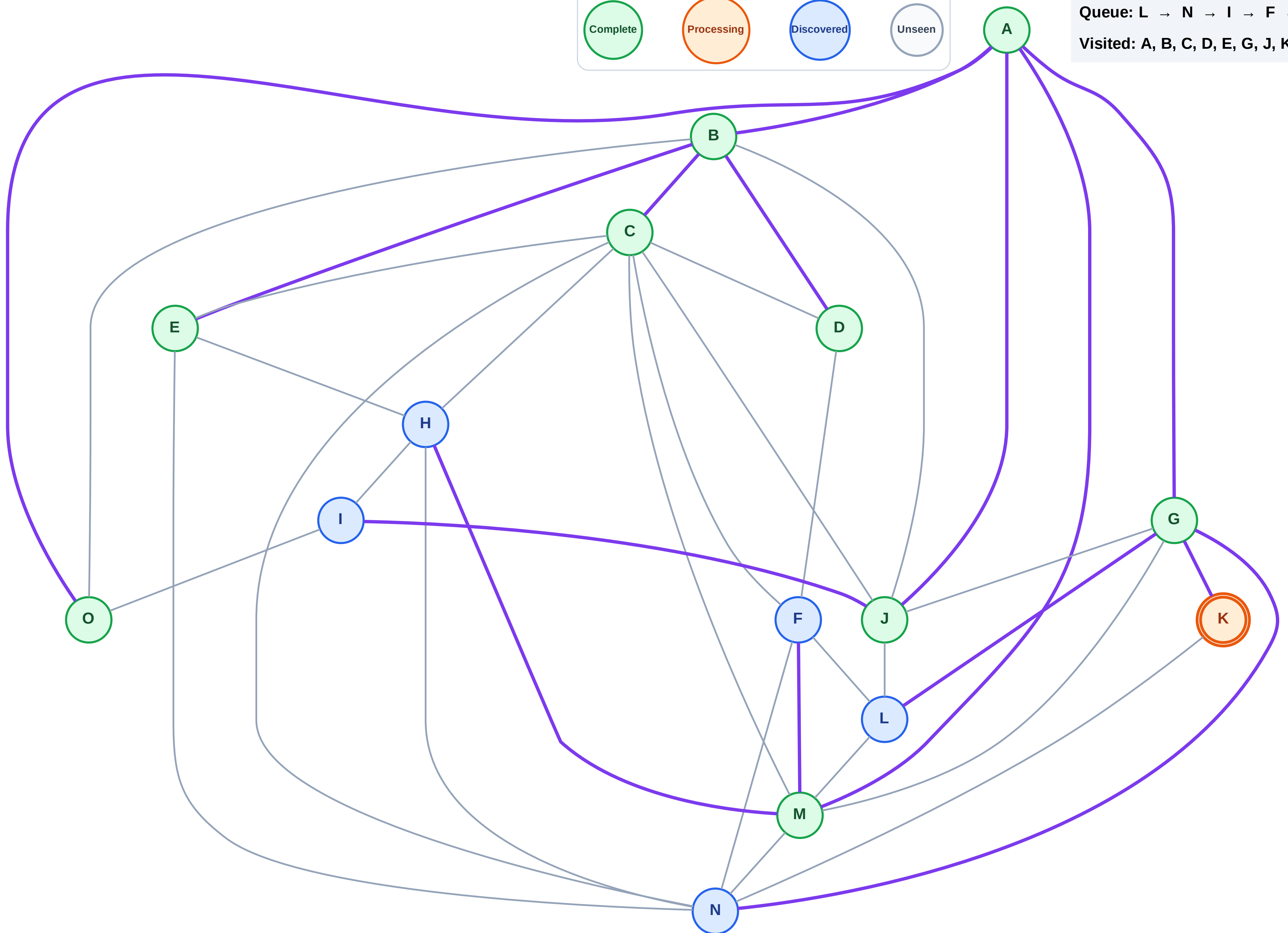
Discovered



Unseen

Queue: L → N → I → F → H

Visited: A, B, C, D, E, G, J, K, M, O



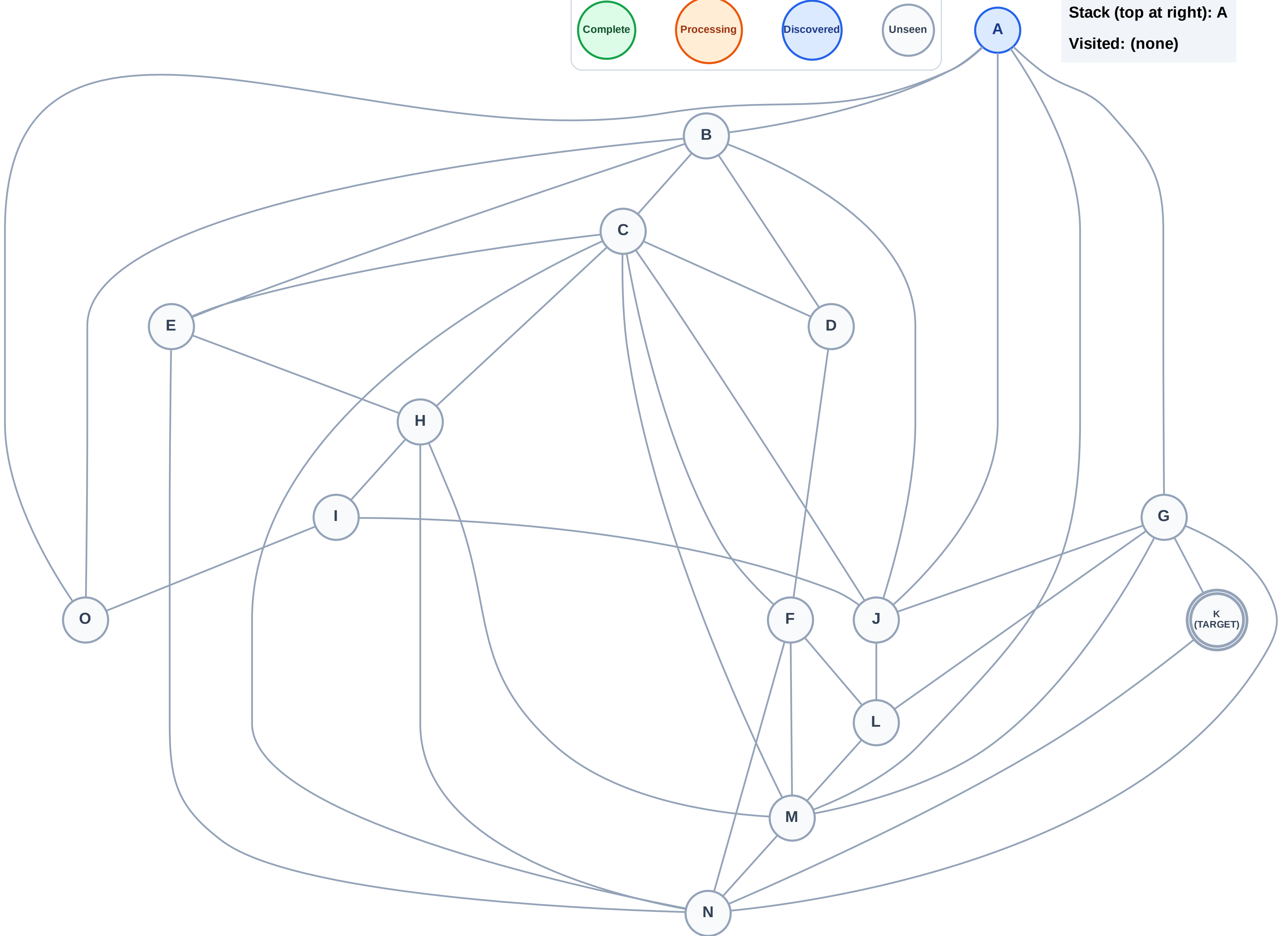
DFS Traversal

Step 1: Start at A (Seeking target node K)

Legend



Stack (top at right): A
Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

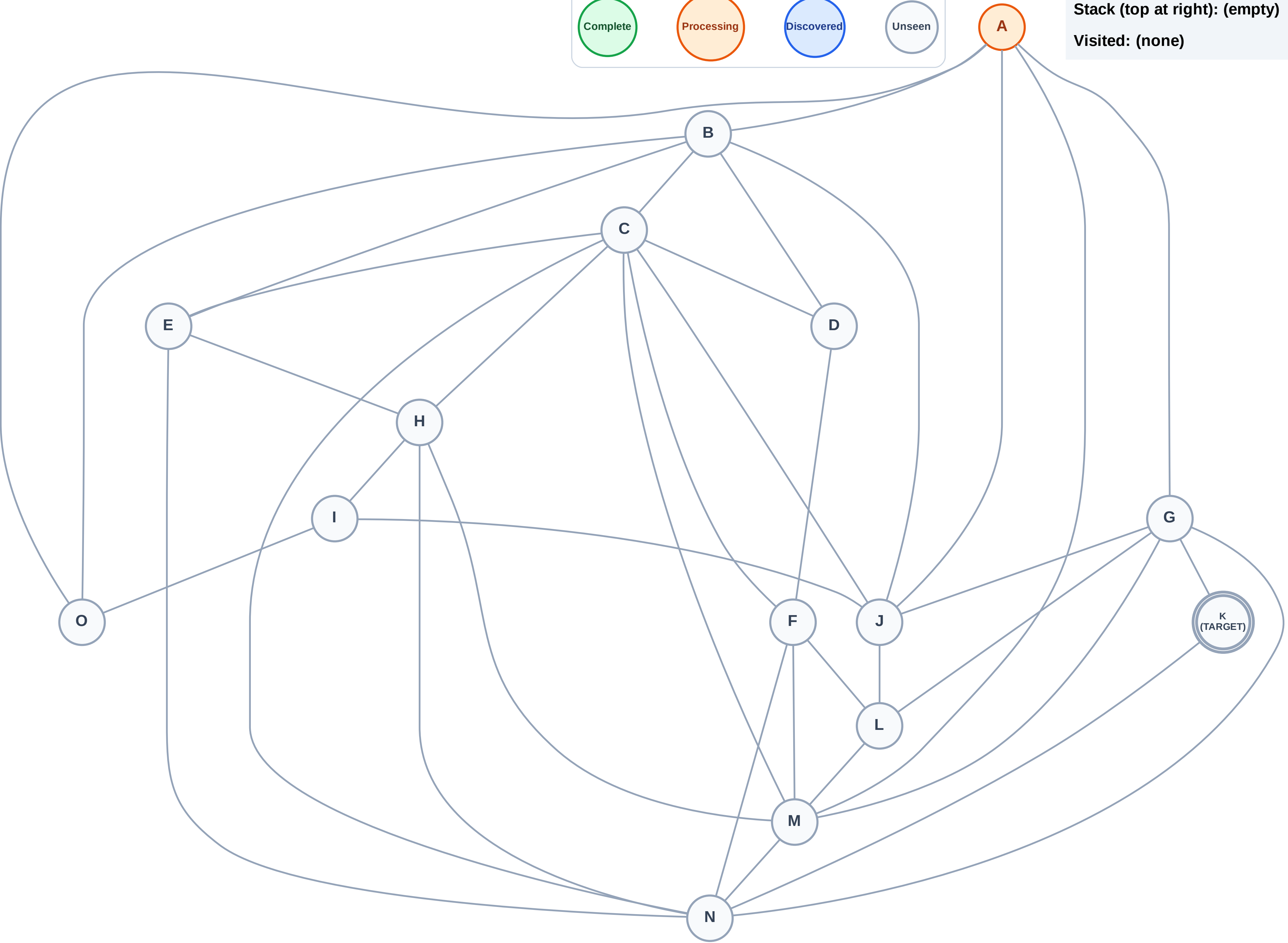
Complete

Processing

Discovered

Unseen

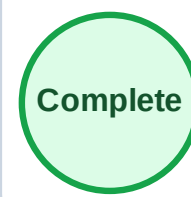
Stack (top at right): (empty)
Visited: (none)



DFS Traversal

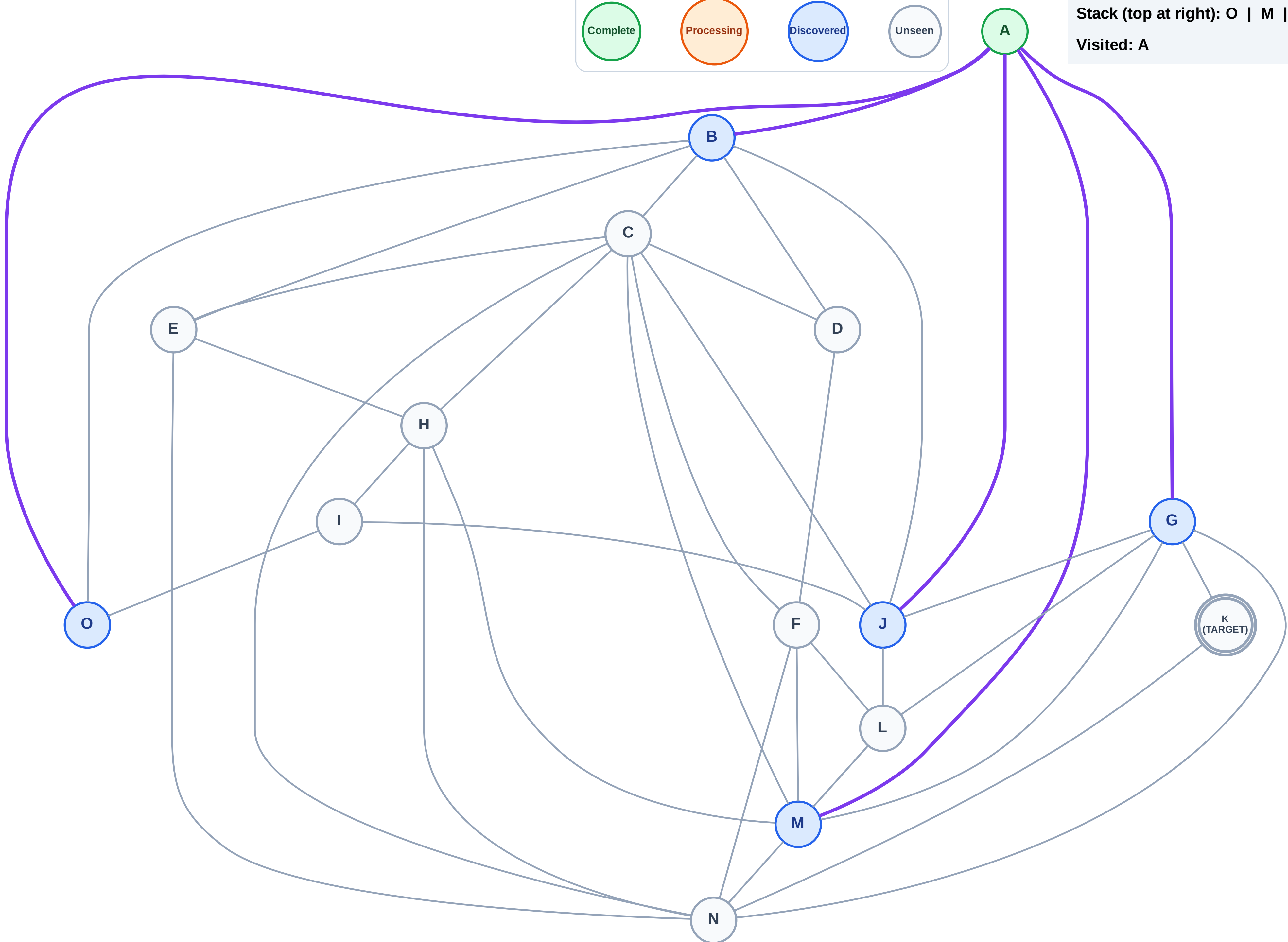
Step 3: Discover O, M, J, G, B from A

Legend



Stack (top at right): O | M | J | G | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend



Complete



Processing



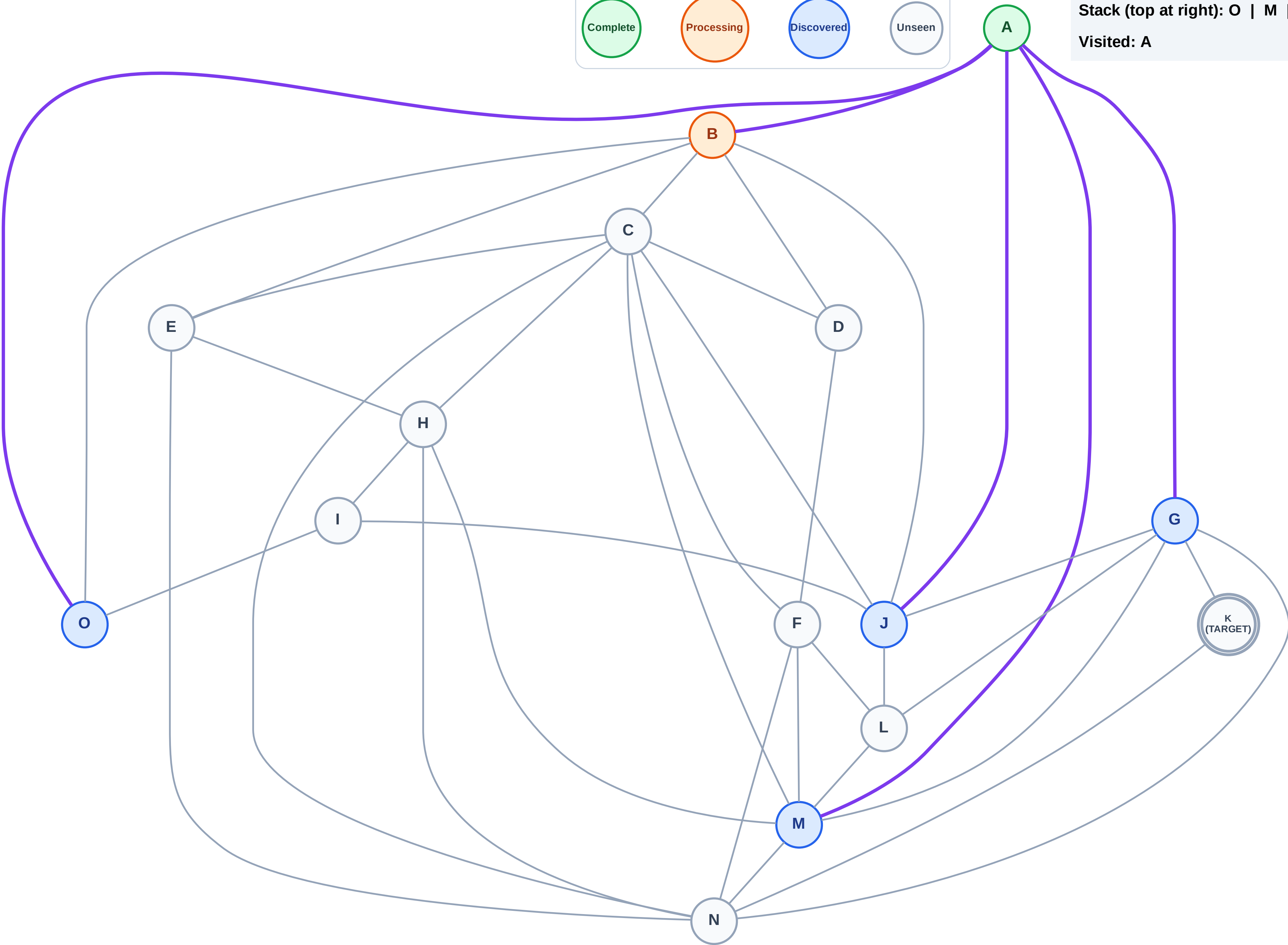
Discovered



Unseen

Stack (top at right): O | M | J | G

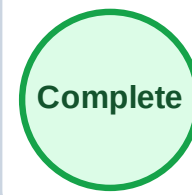
Visited: A



DFS Traversal

Step 5: Discover E, D, C from B

Legend



Complete



Processing



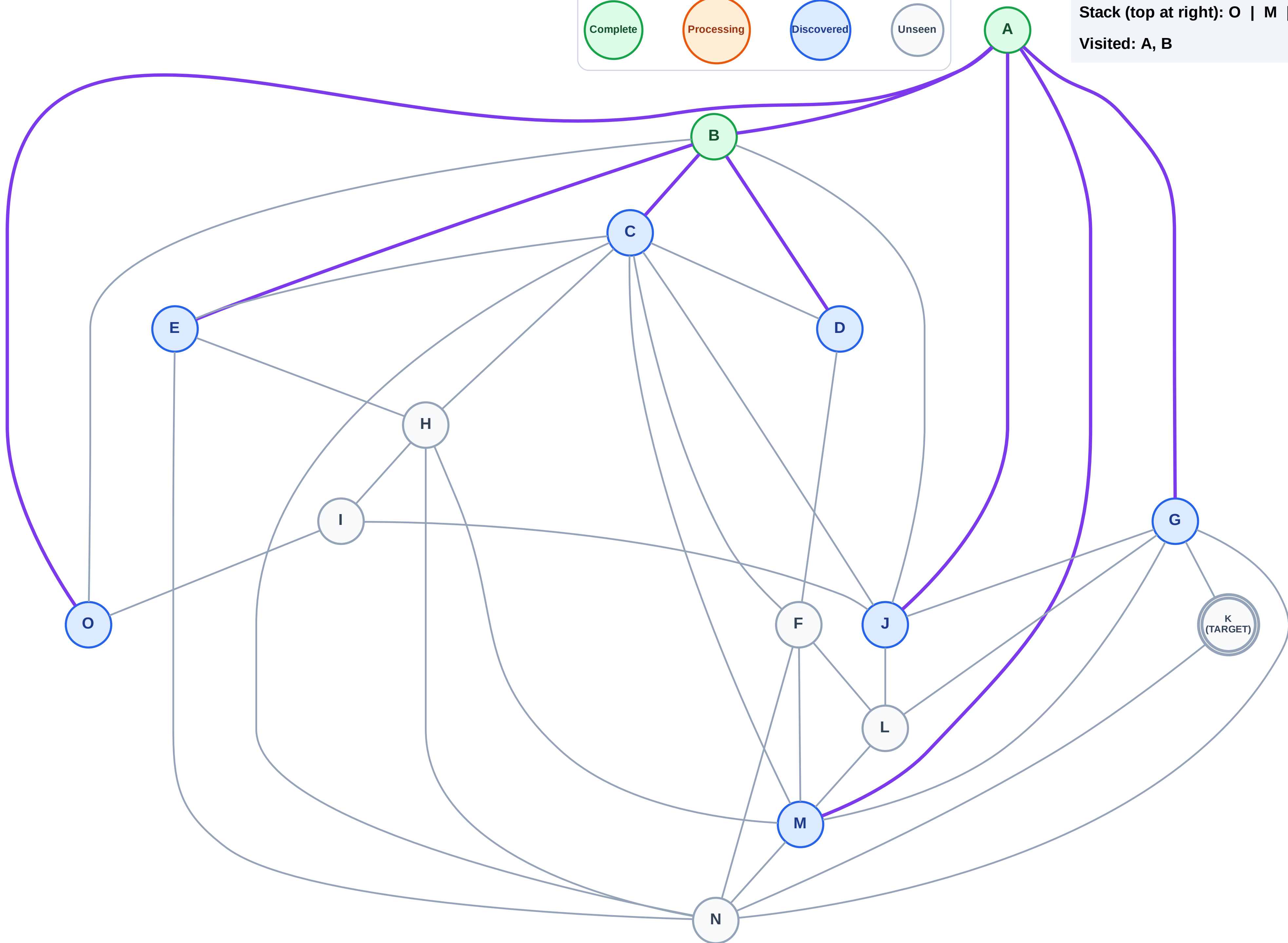
Discovered



Unseen

Stack (top at right): O | M | J | G | E | D | C

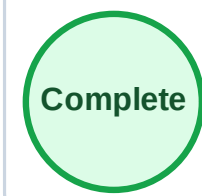
Visited: A, B



DFS Traversal

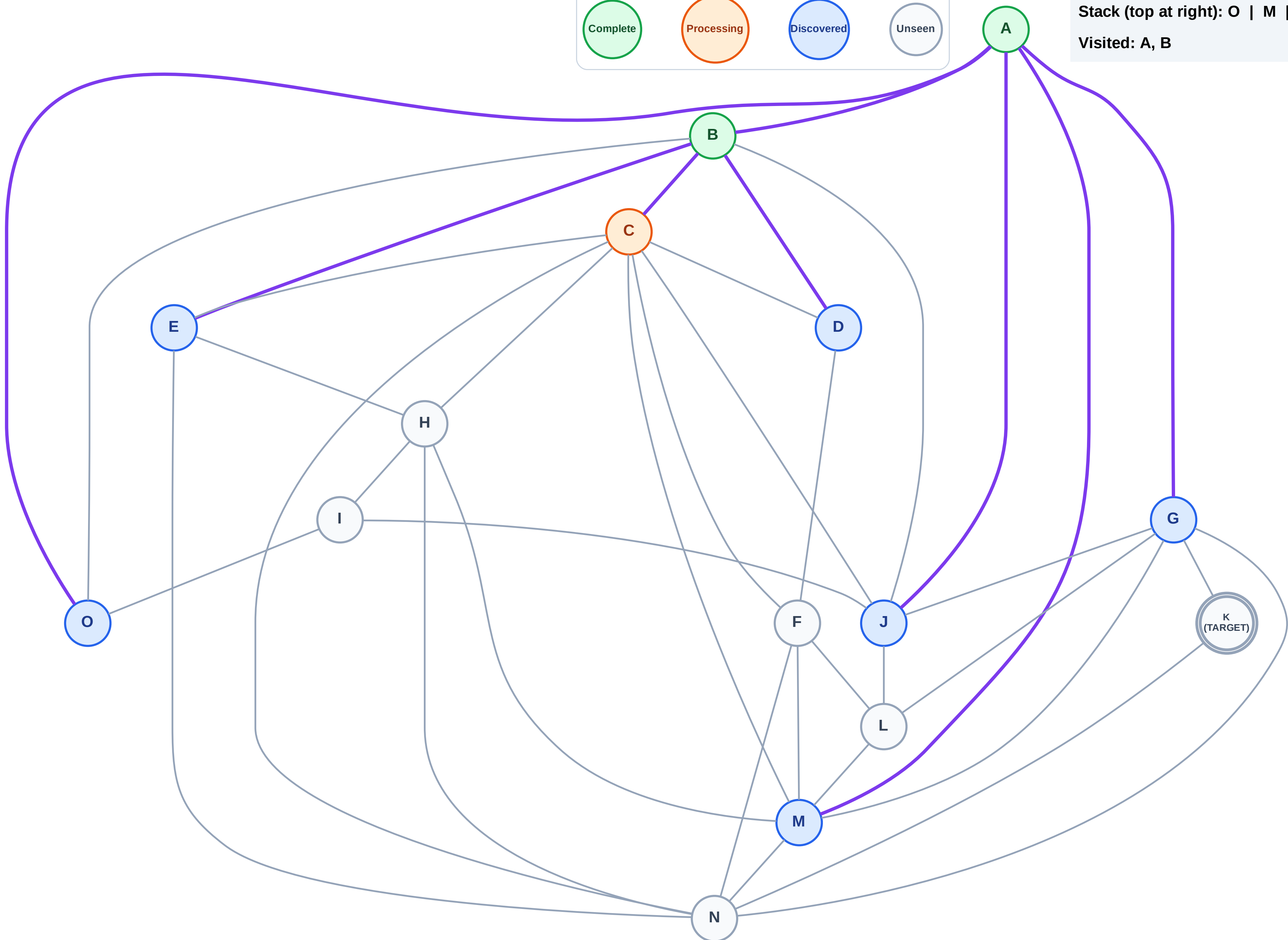
Step 6: Process node C

Legend



Stack (top at right): O | M | J | G | E | D

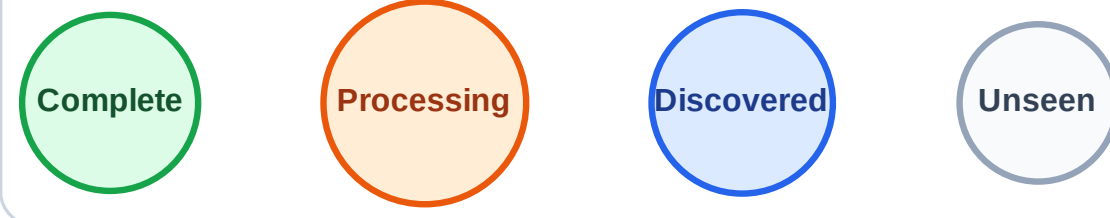
Visited: A, B



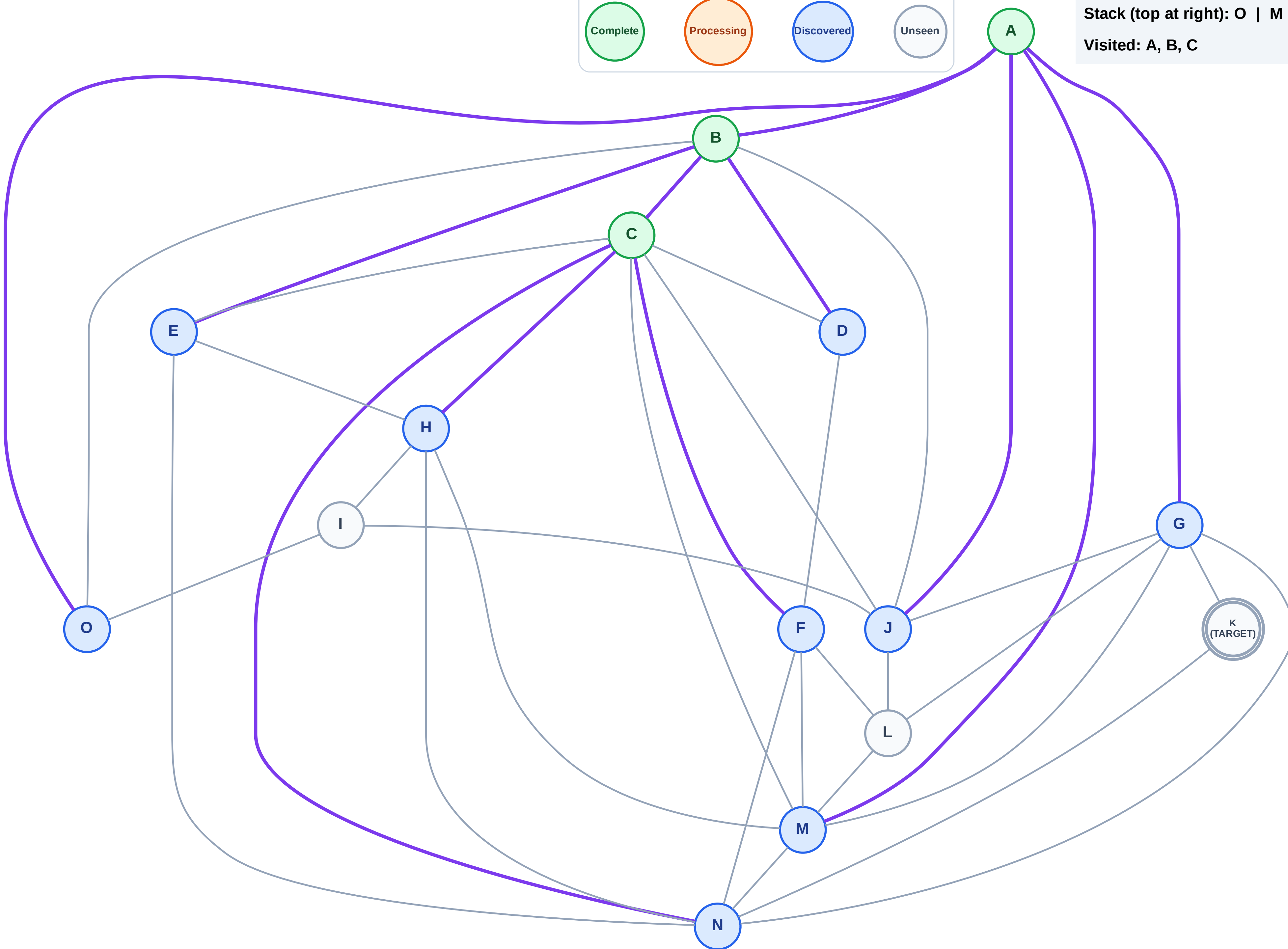
DFS Traversal

Step 7: Discover N, H, F from C

Legend



Stack (top at right): O | M | J | G | E | D | N | H | F
Visited: A, B, C



DFS Traversal
Step 8: Process node F

Legend

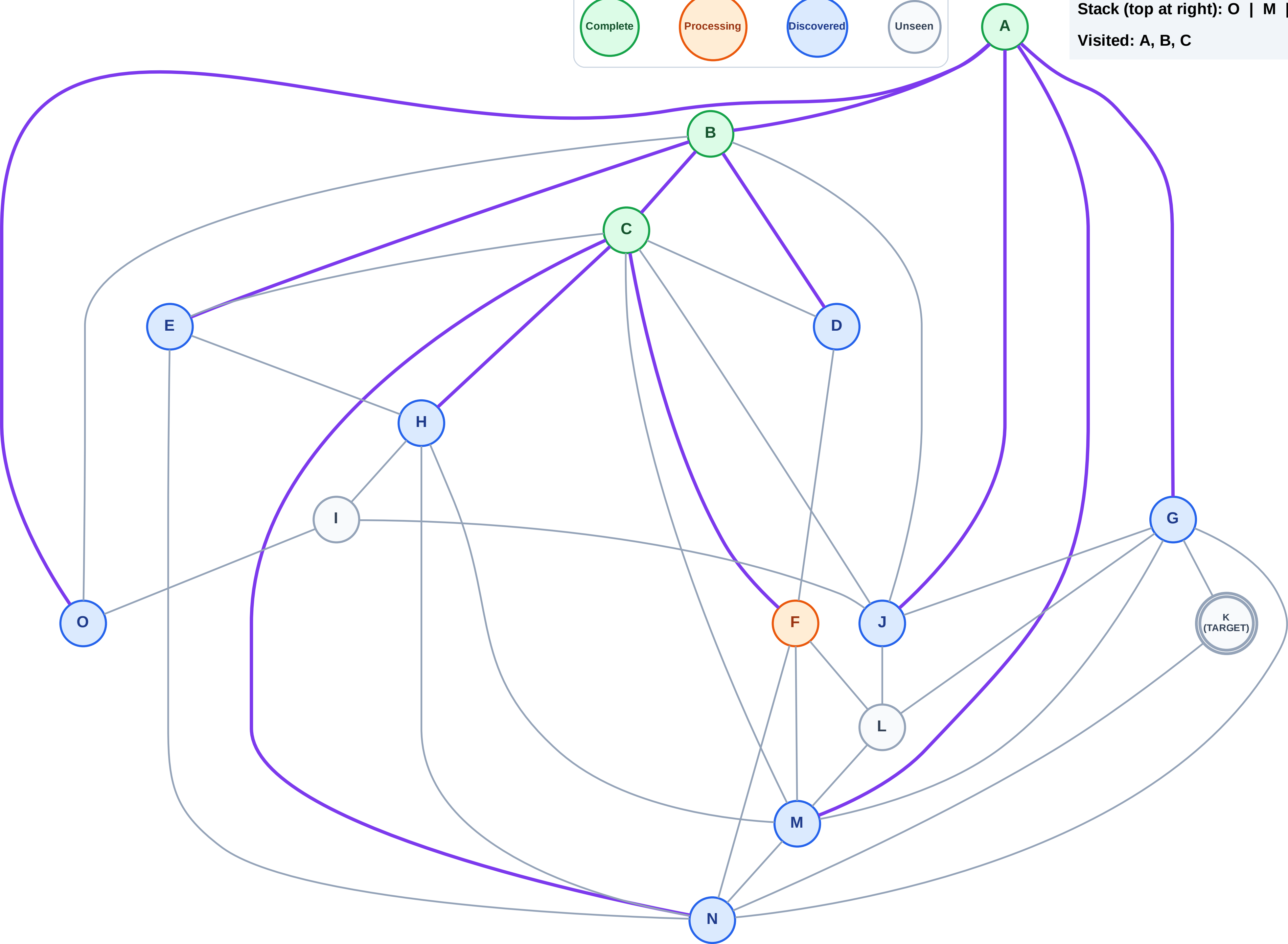
Complete

Processing

Discovered

Unseen

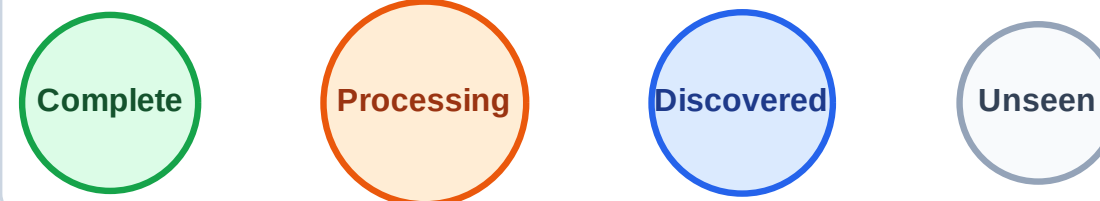
Stack (top at right): O | M | J | G | E | D | N | H
Visited: A, B, C



DFS Traversal

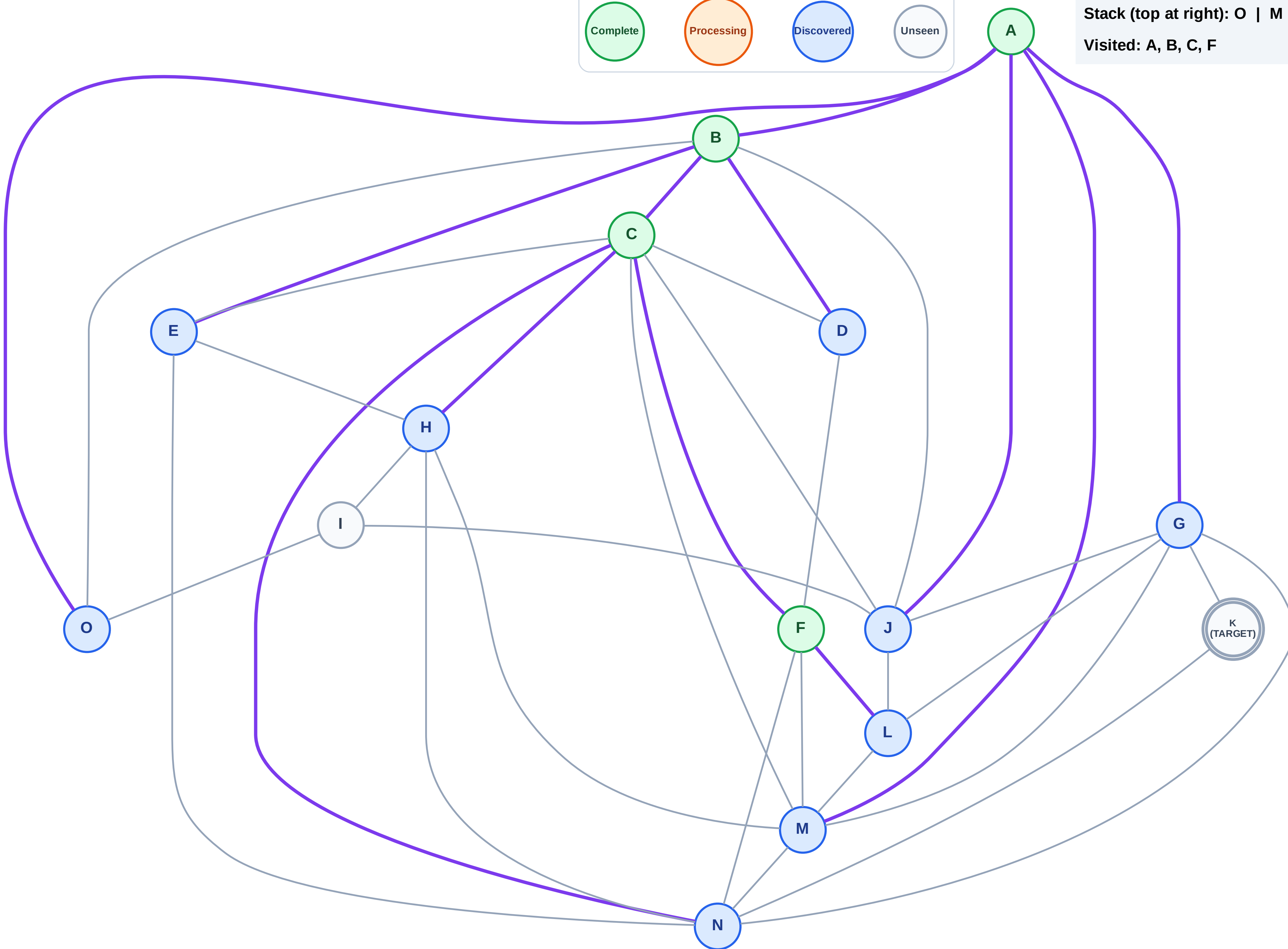
Step 9: Discover L from F

Legend



Stack (top at right): O | M | J | G | E | D | N | H | L

Visited: A, B, C, F



DFS Traversal

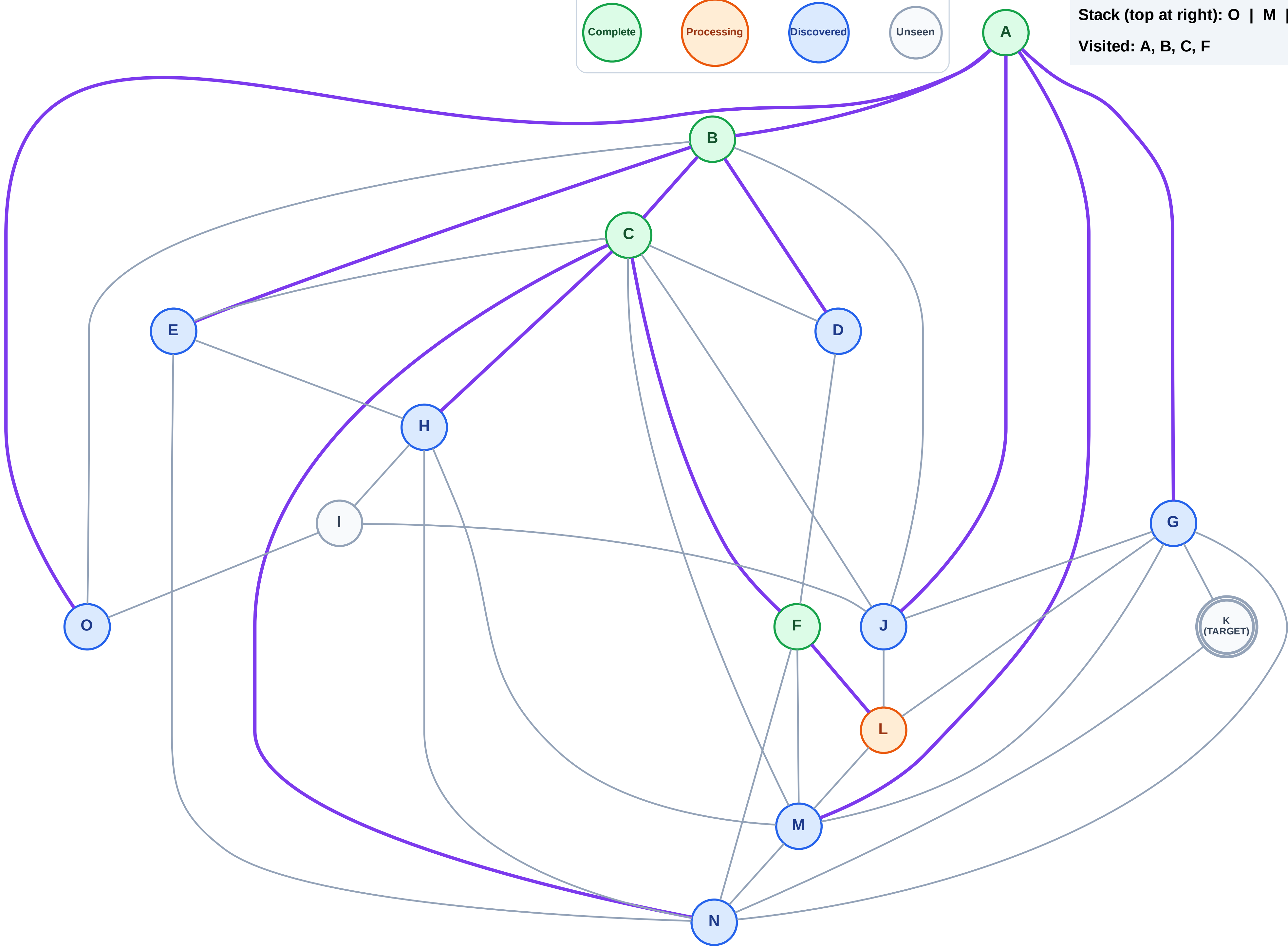
Step 10: Process node L

Legend



Stack (top at right): O | M | J | G | E | D | N | H

Visited: A, B, C, F



DFS Traversal

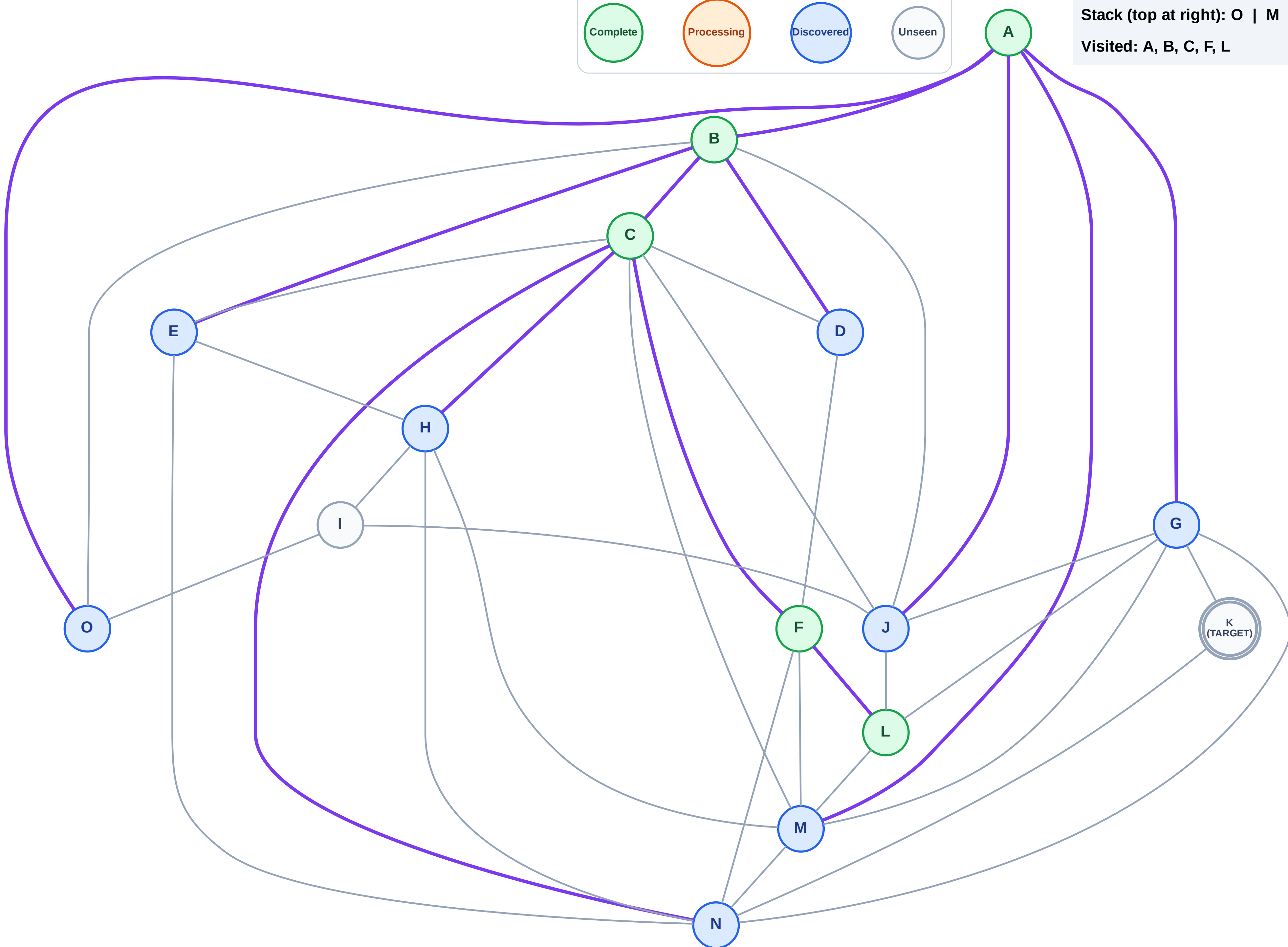
Step 11: No new neighbors from L

Legend



Stack (top at right): O | M | J | G | E | D | N | H

Visited: A, B, C, F, L



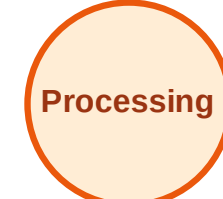
DFS Traversal

Step 12: Process node H

Legend



Complete



Processing



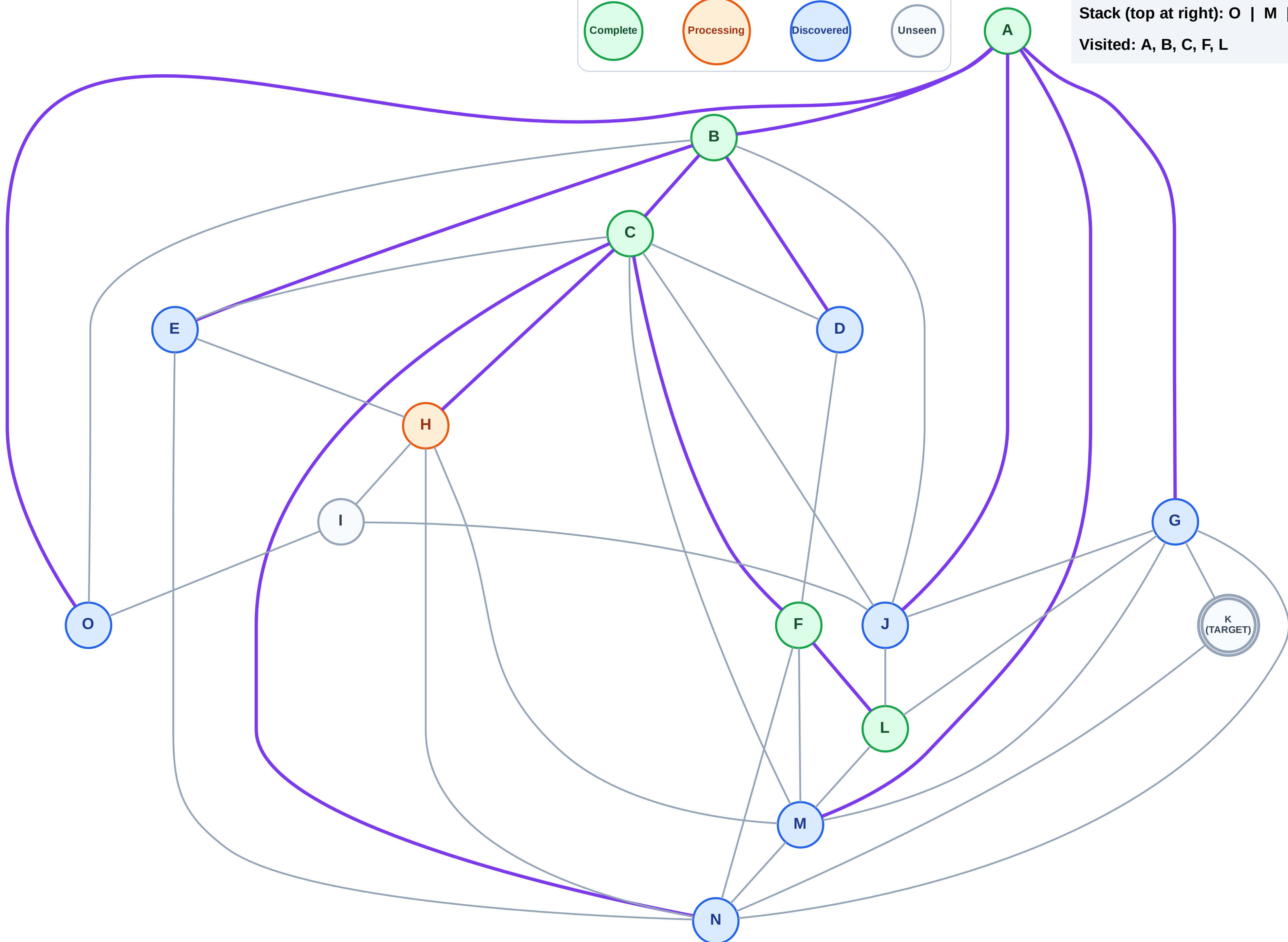
Discovered



Unseen

Stack (top at right): O | M | J | G | E | D | N

Visited: A, B, C, F, L



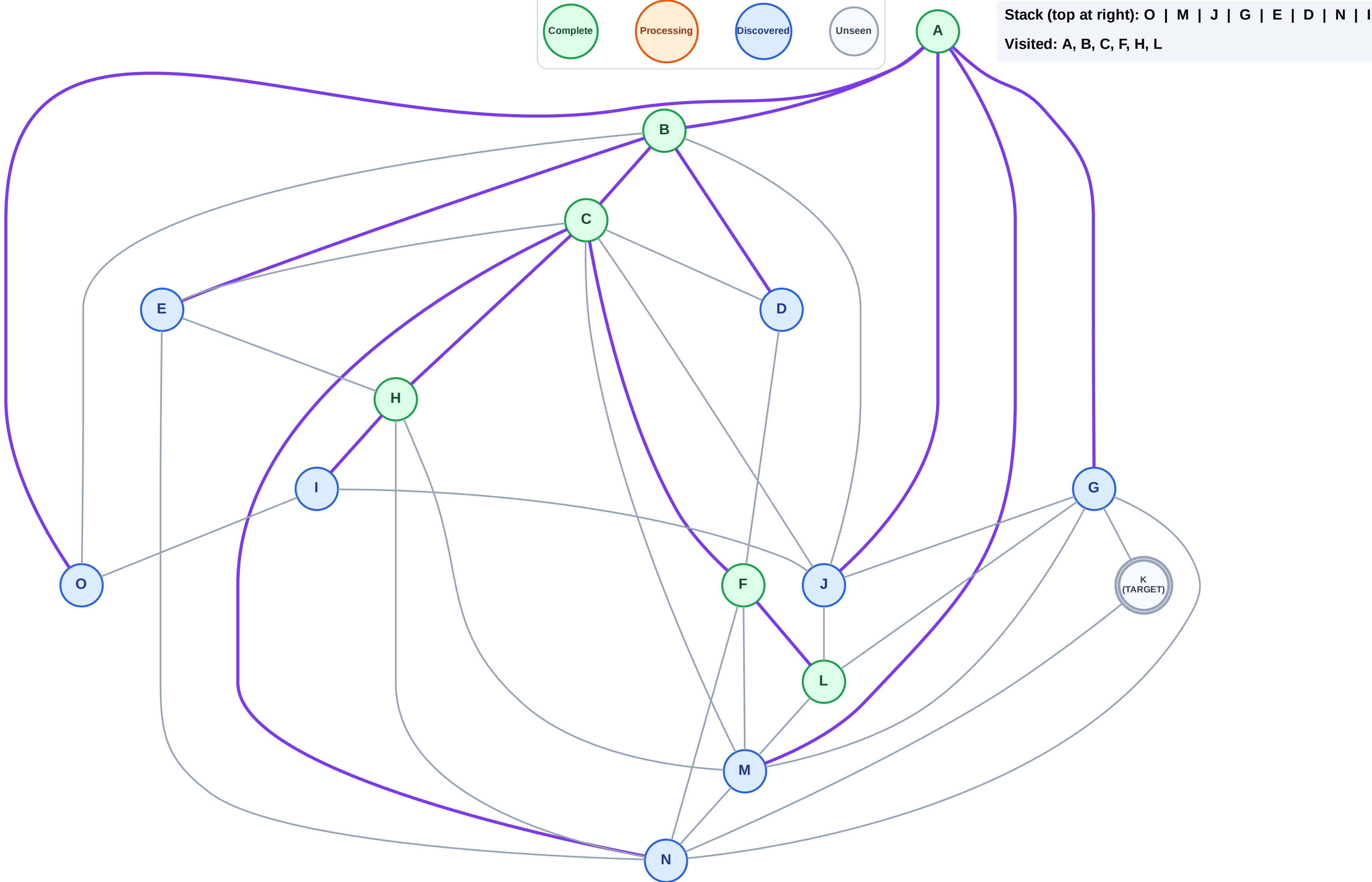
Step 13: Discover I from H

1

...

10

—



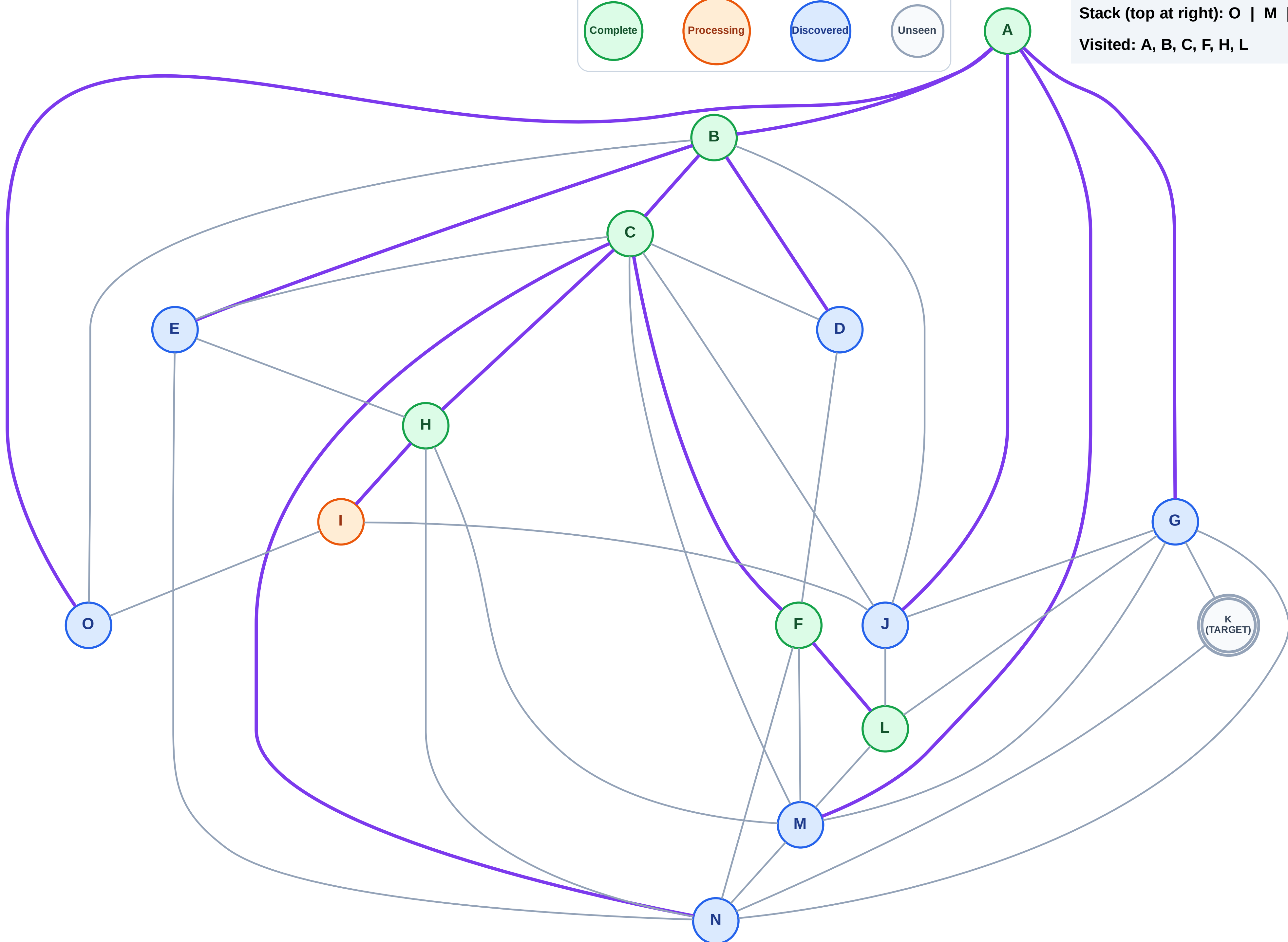
Step 14: Process node I

Legend

Complete Processing Discovered Unseen

Unseen

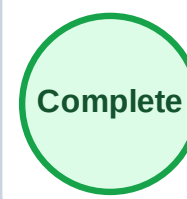
Visited: A, B, C, F, H, L



DFS Traversal

Step 15: No new neighbors from I

Legend



Complete



Processing



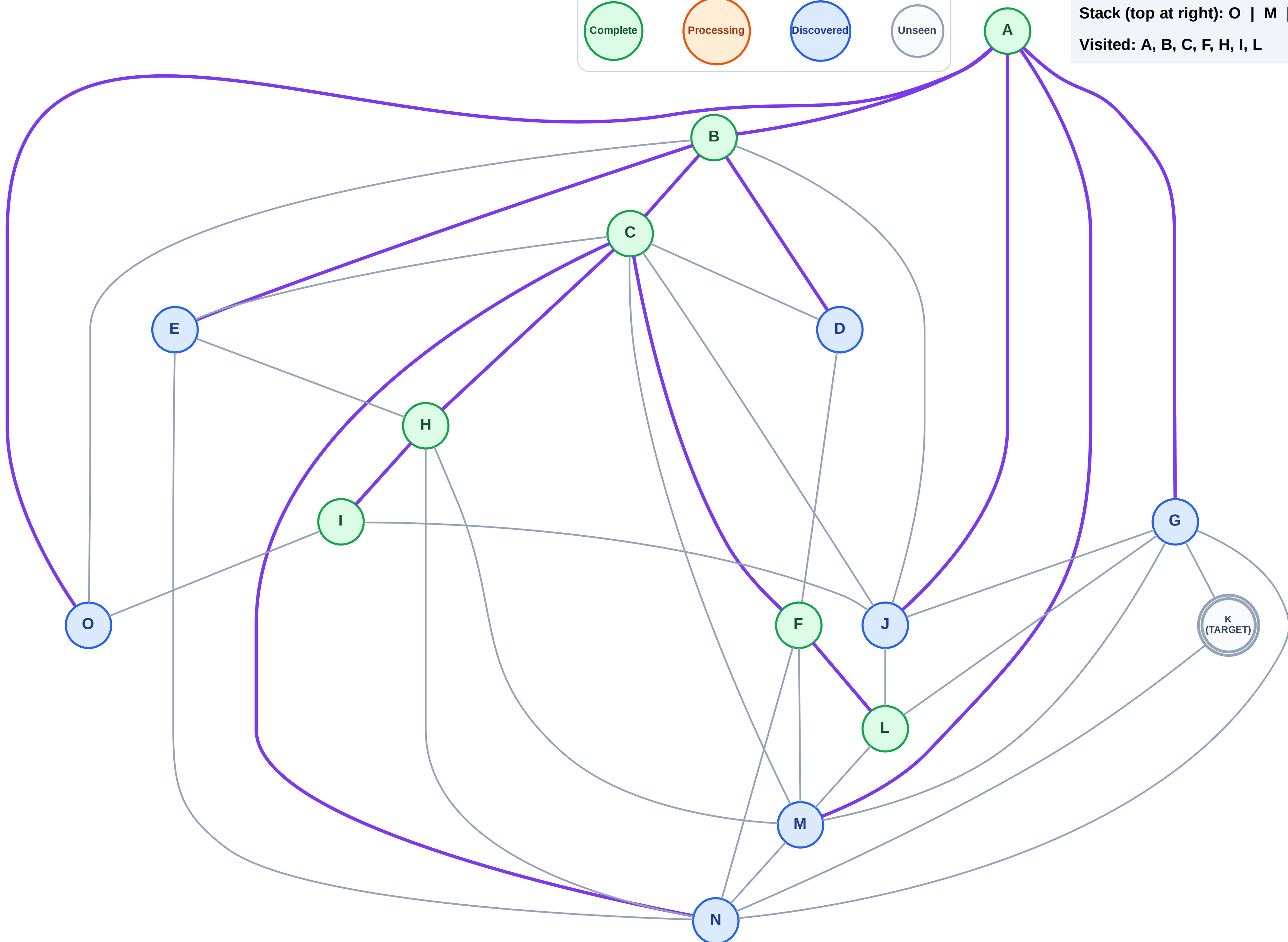
Discovered



Unseen

Stack (top at right): O | M | J | G | E | D | N

Visited: A, B, C, F, H, I, L



DFS Traversal

Step 16: Process node N

Legend



Complete



Processing

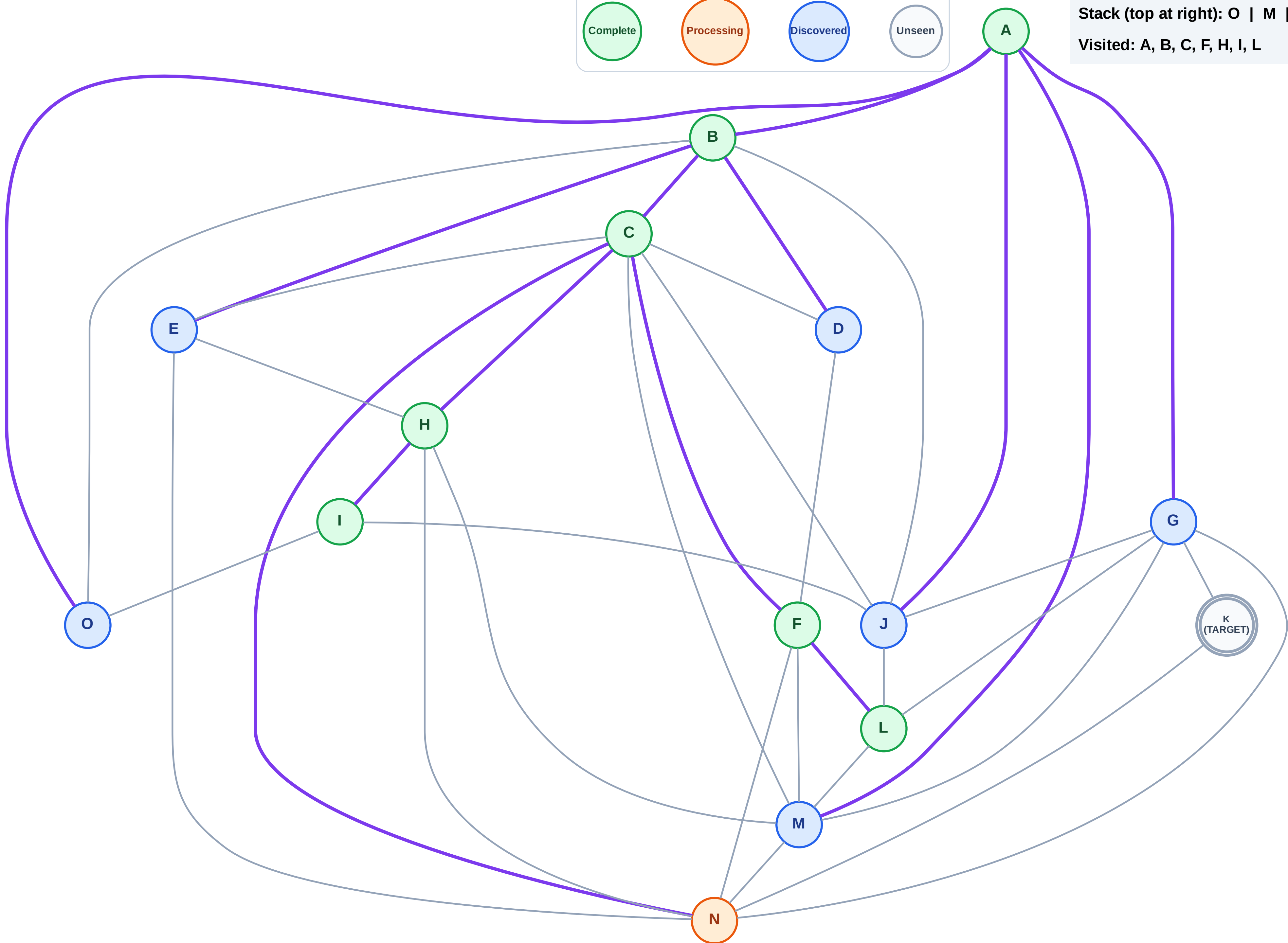


Discovered



Unseen

Stack (top at right): O | M | J | G | E | D
Visited: A, B, C, F, H, I, L



DFS Traversal

Step 17: Discover K from N

Legend



Complete



Processing



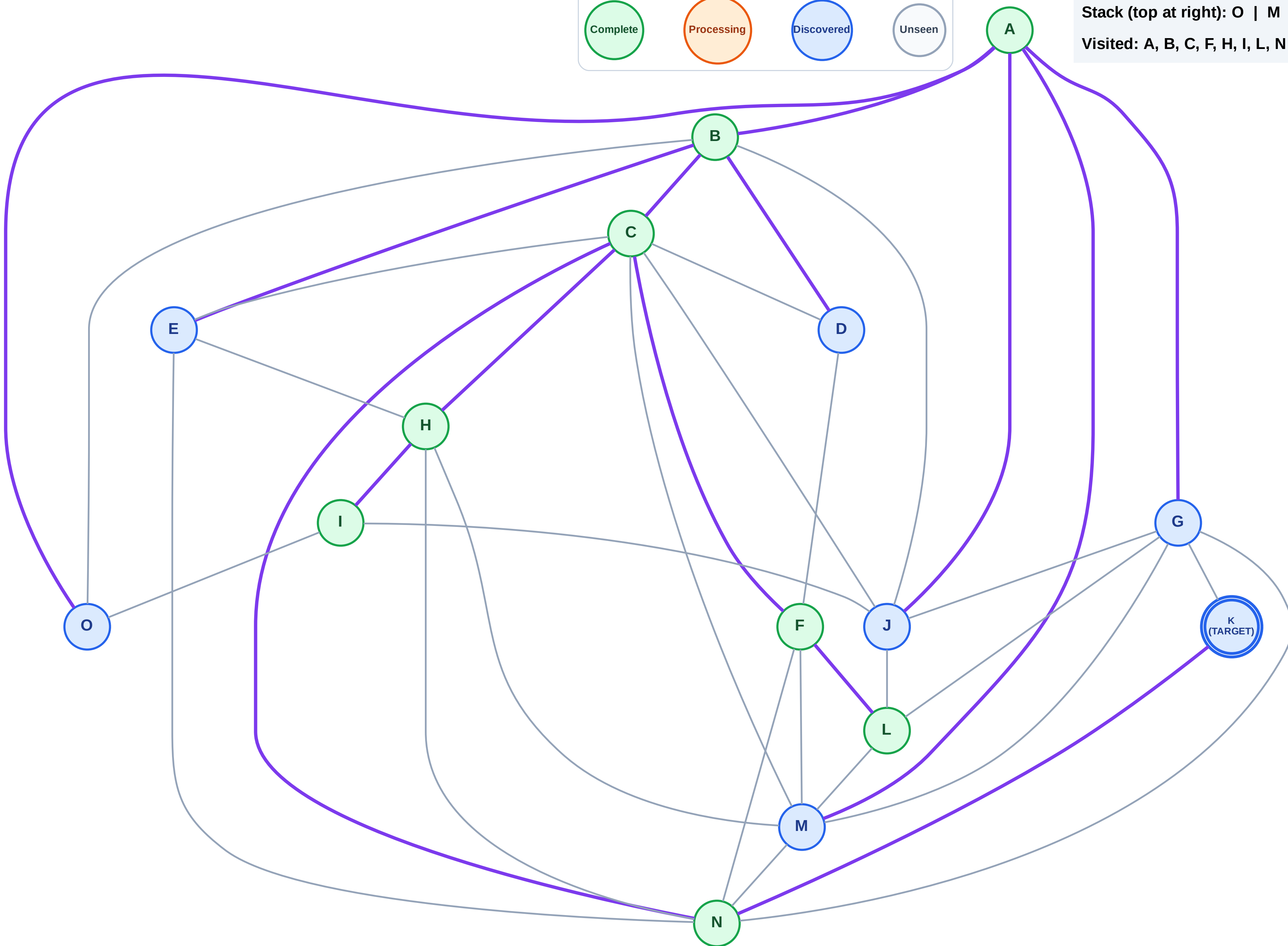
Discovered



Unseen

Stack (top at right): O | M | J | G | E | D | K

Visited: A, B, C, F, H, I, L, N



DFS Traversal

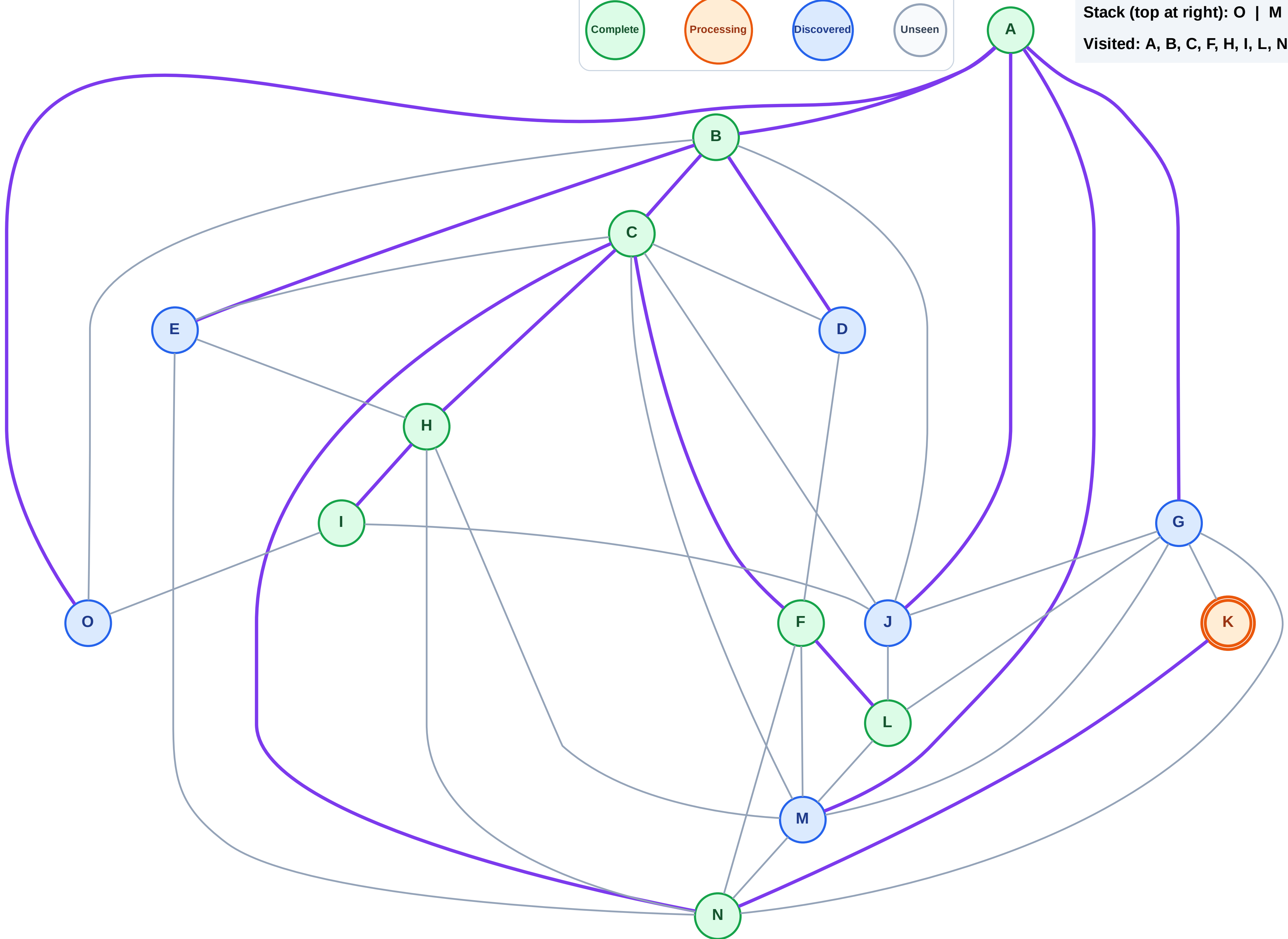
Step 18: Process node K

Legend



Stack (top at right): O | M | J | G | E | D

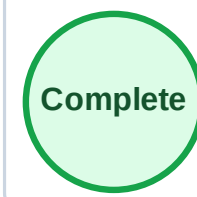
Visited: A, B, C, F, H, I, L, N



DFS Traversal

Step 19: TARGET NODE K REACHED! Path (4 hops): A → B → C → N → K

Legend



Stack (top at right): O | M | J | G | E | D
Visited: A, B, C, F, H, I, K, L, N

